Online Appendix

What Connects Local Housing Markets? Regulation, Migration and the National Cycle

Oğuz Yurtoğlu*

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This Online Appendix collects the robustness checks, complete tables and descriptive exhibits behind the paper. Sections, tables and figures here carry the prefix OA; a reference without it points to the paper or its printed appendix. Where an exhibit has a factor-adjusted version, it sits beside the raw one in the same table. Units and inference follow the paper (Section 4.4), and every table note states its own.

*Koç University, Department of Economics. Email: oyurtoglu22@ku.edu.tr

OA.1 Data Construction, Continued

This section continues Appendix A of the paper, which describes the house-price series, with the migration flows and exposure, the geography and economic-size variables, the national price overlay and the aggregation of the housing-constraint measures.

OA.1.1 Migration flows and exposure

For the annual network of year t, I sum five annual IRS flow matrices from $t-15$ through $t-11$, using IRS interval-start labels. The final migration interval ends before the first year of the benchmark price-response window. If $Q_{ki,t}$ is the pooled flow from origin k to destination i, the origin choice and target feeder shares are

$$\mu_t^{k\to i} = \frac{Q_{ki,t}}{\sum_r Q_{kr,t}}, \qquad \varphi_t^{i\leftarrow k} = \frac{Q_{ki,t}}{\sum_r Q_{ri,t}}. \tag{OA.1}$$

Observed same-state exemption counts remain on the diagonal. Both denominators therefore use the full 48-state system. Exposure is $\Psi_t = \Phi_t M_t$, and only after the complete row-stochastic matrix is formed do the regressions remove its diagonal cells. Exposure is measured in percentage points, $100\Psi_t$, and its diagonal in unit-share form gives external openness $o_{i,t} = 1 - \psi_{ii,t}$. Because the stayers are on the diagonal, the direct terms of equation (8) dominate the off-diagonal cells, and the common-origin term averages 1.3 percent of external exposure (Table OA.20).

The 2005 network uses IRS interval-start vintages 1990–1994, and the sequence ends with 2010–2014 for the 2025 network. All objects use the same target-row/source-column orientation.

OA.1.2 Geography, economic size and the national price overlay

Interstate distance is the great-circle distance in kilometers between 2000 Census population centers. Contiguity denotes a shared state border, and common region denotes membership in the same Census region. Target average distance is the arithmetic mean of distances to the other 47 states, logged after averaging. The state size control is the logged BEA resident population for year $t-1$, available for every year from 1985. It falls inside the price window and supplies a size control, not pre-window identification, and it may itself respond to regulation. Table OA.5 therefore reports the sending-side regressions without it and with the state's 1980 population from the same BEA series in its place. The 1980 value precedes the end of every window, although it lies inside the windows ending in 1985–1989. The factor-adjusted TO association is 8.0 percent without the control, 7.5 with lagged population and 7.5 with 1980 population. GDP enters only the 2005–2025 variant reported in Online Appendix Section OA.4.1, as the logged BEA state series for year $t-1$, real GDP, all-industry total, in millions of chained 2017 dollars (BEA SAGDP9, line 1; equivalently SASUMMARY, line 1 for overlapping years), which does not reach back before 1997.

The metropolitan size controls come from Census Bureau county population estimates for 1980–2024. They combine the 1980s and 2000s intercensal series, the CO-99-8 estimates for the 1990s, and the Vintage 2020 and Vintage 2024 postcensal series, one source per year, put on the 2020 county geography (Dade to Miami-Dade, the Virginia independent cities that merged into their counties, and Shannon to Oglala Lakota). Connecticut's Vintage 2024 figures are for planning regions, so its counties carry their 2020 shares of the state total from 2020 on, and two corrupted 1994 cells of the CO-99-8 release are interpolated. The county sums reproduce every state total in the releases to within 0.03 percent. A CBSA's population is the sum over its 2020 member counties. The outside-node-set share is the share of the population within 200 kilometers of the CBSA's internal point that lives in counties belonging to no sample area, a county counting as within the radius when its internal point is. Both are for year $t-1$.

The national real-price overlay divides the national (US) column of the same FMHPI file by the BLS seasonally adjusted CPI-U and normalizes the ratio to 100 in January 2000. Like the state network estimates, it runs through December 2025. October 2025 has no published CPI observation, is not interpolated and is drawn as a one-month gap.

OA.1.3 Housing-constraint aggregation

The constraint proxies originate at different geographic units, so a state-level measure requires an explicit aggregation rule. The target-level models use the Lutz–Sand share unavailable because of steep slopes, aggregated from whole metropolitan and micropolitan areas using 1990 total housing stock and the cross-state allocation described below. The primary Wharton indexes are equal-jurisdiction means among responding jurisdictions in each state,

$$\mathrm{WRLURI}_{s,v} = \frac{\sum_{r \in s} w_{r,v} \mathrm{WRLURI}_{r,v}}{\sum_{r \in s} w_{r,v}}, \qquad w_{r,v} = 1, \tag{OA.2}$$

for state s and vintage $v \in \{2006, 2018\}$. Larger values of each proxy denote greater constraint. The Lutz–Sand series and both Wharton vintages cover all 48 contiguous states. The empirical measures are proxies for the model's unobserved inverse supply parameter η_i^H. The jurisdiction survey carries no population to weight by, so no population-weighted WRLURI series is reported. The Wharton inverse-probability survey weights and Census jurisdiction land area produce genuinely different state measures. The Saiz land measure of Table 5 is genuinely population-weighted. It averages the unavailable-land shares of each state's metropolitan areas with 2000 population weights, each area counted in the state named first in its title. The metropolitan index of Section OA.6.1 follows the same equal-jurisdiction rule over the jurisdictions of each metropolitan area. State coverage means that a value is available for every state, not that responding jurisdictions represent its entire housing market. Equal-jurisdiction means may therefore reflect uneven survey coverage. The unchanged 2006 and 2018 state averages have a cross-state correlation of 0.561. Their unstandardized state-index standard deviations are 0.710538 and 0.422642, respectively; each vintage is standardized over the 48 states. This compares survey vintages as well as potentially different sets of responding jurisdictions;

the 2018 survey also falls within the estimation period rather than preceding all of it. Table 5 reports the vintage comparison and Online Appendix Table OA.21 the aggregation checks.

Land constraints over local housing markets. The benchmark and alternative land measures retain each Lutz–Sand core-based statistical area (CBSA) as a whole, including its outlying counties. The 2020 county-defined geography contains both metropolitan and micropolitan areas. To allocate a market that crosses state borders, let $n_{ms}^{b,v}$ be population or total housing stock (basis b) in the counties of CBSA m that lie in state s, measured in Census year $v \in \{1990, 2000\}$. Let $n_m^{b,v} = \sum_s n_{ms}^{b,v}$. The cross-state allocation share and normalized within-state weight are

$$a_{ms}^{b,v} = \frac{n_{ms}^{b,v}}{n_m^{b,v}}, \qquad \omega_{ms}^{b,v} = \frac{a_{ms}^{b,v} n_m^{b,v}}{\sum_k a_{ks}^{b,v} n_k^{b,v}} = \frac{n_{ms}^{b,v}}{\sum_k n_{ks}^{b,v}}. \tag{OA.3}$$

The state proxy is $\sum_m \omega_{ms}^{b,v} \ell_m$, where ℓ_m is the whole-CBSA share unavailable because of steep slopes. A CBSA entirely within a state has allocation share one; a cross-border CBSA contributes according to that state's share of its population or housing stock. This allocation is applied once, before the within-state weights are normalized. The benchmark uses total housing stock in 1990. All four combinations of weighting basis and Census year are reported.

These measures describe constraints over covered local housing markets, including land outside their inhabited centers, rather than over every county in the state. They use the released unbuffered CBSA polygons. The supplied positive-buffer fields duplicate those values and do not establish an additional surrounding-land measure. The 2020 boundaries provide a common geographic definition, not historical market boundaries for 1990 or 2000. Two historical boundary changes need a convention. Broomfield County, Colorado, was created after both weighting years from parts of four counties, so the Denver, Boulder and Greeley markets are combined into one, with the combined steep land share an area-weighted mean of the three released shares; and the 46 housing units of the former Yellowstone county equivalent in Montana, whose locations in 1990 are not observed, are assigned to Park County. Under the alternative convention for each, land-area-only weights for Colorado and Gallatin County for Montana, the regression coefficients are unchanged to two decimals.

All four weighting rules, both boundary conventions and the county slope comparator, which uses the unbuffered county steep land shares and 1990 one-unit housing-stock weights, cover the same 48 states, and each measure is standardized over them. Across the seven, the steep land share coefficient lies between 1.1 and 1.6 log points for raw TO and between -0.9 and -0.6 for factor-adjusted TO, with standard errors above 2.3 in every case, so no physical-constraint estimate is distinguishable from zero at the 5-percent level, and the WRLURI coefficient moves within 0.1 log points (Section OA.8).

OA.2 Networks and Connectedness Indices

OA.2.1 Testing the decline in the distance gradient

The per-endpoint regressions behind Figure 3 each carry their own covariance matrix, so the fall in the gradient cannot be tested from them. The same regressions fitted jointly—log distance interacted with endpoint, with source-by-endpoint and target-by-endpoint effects—leave every point estimate numerically unchanged, because fully interacted node effects make the Poisson likelihood separable by endpoint. At twelve endpoints the two sets of estimates agree to machine precision. What the joint fit adds is one covariance matrix, and with it a Wald test. The mean gradient over 2015–2025 is 0.301 below the mean over 1990–1999, with a two-way clustered standard error of 0.045. Two cautions attach. The clustering has 48 source and 48 target states, so the effective number of clusters is small relative to the 252 interacted coefficients, and the joint covariance is rank-deficient by construction; the contrast tested here is one-dimensional and needs only its own variance, but a joint test across the coefficients would not be supported.

The figure uses the row-normalized cell, and the gradient on unnormalized cells is a different estimand. A log link with target-by-endpoint effects absorbs a constant common to every row exactly, but the row total varies across rows, and Poisson quasi-likelihood sets the group effect so that fitted and observed totals agree within the group, so a row rescaled by its own constant contributes that constant times as much to the slope score. The fixed effect absorbs the level, not the weight. The row weights can be equalized without the unnormalized cells. Re-estimating every gradient on the conditional shares of equation (12), which sum to 100 in every target row and so give every target the same weight, leaves the series correlated with the drawn ones at 0.999 in the raw state network, 0.996 in the factor-adjusted one and 1.000 and 0.999 at the metropolitan scale, with no endpoint differing by more than 0.031. The raw state gradient still falls, from 0.408 in the 1990s to 0.118 during 2020–2025, while the factor-adjusted one averages 0.159 over the annual networks and has no trend distinguishable from zero. The contrast between the two networks does not come from how targets are weighted.

OA.2.2 Metropolitan connectedness over the national cycle

Figure OA.1 repeats Figure 1 for the 100 metropolitan areas. The raw index rises from an average of 87.2 over the annual networks of 1985–1989 to 96.2 since 2020, and the factor-adjusted index averages 87.4 and 85.7 over the same endpoints, 86.2 across all monthly endpoints, and stays between 81.8 and 89.1. As at state scale, the two are weakly negatively correlated month by month, -0.33, so the rise of metropolitan connectedness also runs through the national factor. The factor-adjusted indexes of the two systems move together, with levels that correlate at 0.83 across the monthly endpoints.

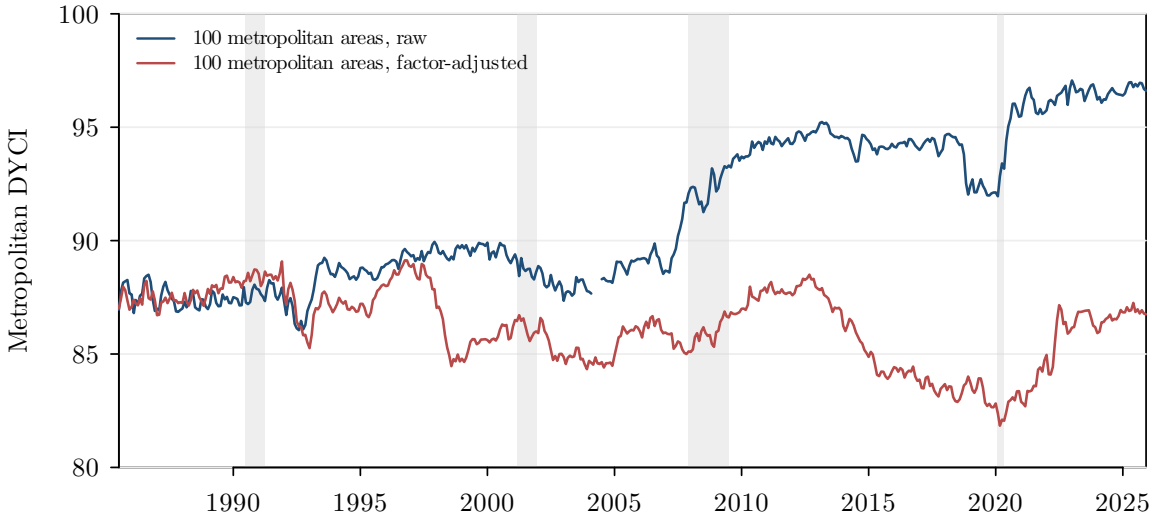


Figure OA.1: Metropolitan connectedness over the national housing cycle, raw and factor-adjusted, July 1985–December 2025

Note: Monthly Diebold–Yilmaz Connectedness Index of the 100-metropolitan-area system from the 120-month rolling elastic-net VAR(2) at the benchmark specification of Section OA.3.4, in navy for the raw network and in red for the factor-adjusted one, estimated after each window's first principal component of the 100 areas' monthly growth is removed (Section 2.3). The four raw endpoints March–June 2004, whose fitted companion matrix has an eigenvalue outside the unit circle, have no variance decomposition and are drawn as gaps, and the factor-adjusted network is stable at all 486 endpoints. Levels are not comparable with those of the 48-state system, which has 47 external sources per node against 99 (Section OA.3.4). Shaded areas denote NBER recessions.

OA.2.3 Lag order and the information criterion

Table OA.1 reports the Bayesian information criterion behind the lag order of Appendix B.1, computed on the benchmark estimator itself with one, two and three lags of every state. With the penalized fit's nonzero coefficients as degrees of freedom, two lags has the lowest criterion in 40 of the 41 annual networks. With each equation refitted by least squares on its retained slopes, one lag is still rejected by a wide margin, but three lags has a slightly lower criterion than two at 25 of the 38 endpoints at which every equation can be refitted, so on that count the criterion does not separate two lags from three. At three endpoints one three-lag equation retains more slopes than it has observations and cannot be refitted.

Table OA.1: Lag order and the Bayesian information criterion at the benchmark estimator, 48 states

Lags K	Active slopes per equation	Penalized fit			Post-selection refit	
		BIC	Difference from $K = 2$	Endpoints preferred	Difference from $K = 2$	Endpoints preferred
Panel A. The 41 annual networks 1985–2025 (38 in the post-selection columns)						
1	13.8	−70,337	+857	0	+2,005	0
2	25.1	−71,194	–	40	–	13
3	25.7	−70,764	+430	1	−118	25
Panel B. The 21 annual networks 2005–2025 (19 in the post-selection columns)						
1	14.5	−71,748	+921	0	+2,197	0
2	27.0	−72,669	–	21	–	6
3	28.2	−72,183	+486	0	−106	13

Note: The 48-state system. In each annual network the benchmark estimator of Section 2 is refitted with $K = 1$, 2 and 3 lags of every state and nothing else changed. Each equation is an elastic net with a mixing weight of 0.5 on the same 120 response months, with pre-window months supplying the lags, and its penalty is the one that minimizes hv-block validation loss over twelve contiguous ten-month blocks, with a buffer of K months on either side of each block. Own lags carry the cross-lag penalty. For equation e the criterion is $n \log(\mathrm{RSS}_e/n) + \mathrm{df}_e \log n$ with $n = 120$. In the penalized columns RSS_e is the in-sample residual sum of squares at the selected penalty and df_e is the number of nonzero slopes plus the intercept, the usual effective degrees of freedom of a lasso-type fit, exact for the lasso and an approximation for the elastic net. In the post-selection columns each equation is refitted by least squares on the slopes the penalty retains, with the same count. At three endpoints (1999, 2014 and 2021) one three-lag equation retains 120 or more slopes and cannot be refitted, so the post-selection columns use the endpoints at which every equation of every lag order can be. The system criterion is the sum over the 48 equations. BIC is its mean over the endpoints, and a difference is the mean of the endpoint-by-endpoint differences from $K = 2$, so a positive difference favors two lags. Endpoints preferred counts the endpoints at which a lag order has the lowest criterion. Active slopes are the nonzero lag coefficients per equation, own lags included. The $K = 2$ fits reproduce the stored benchmark network at every endpoint.

OA.3 Estimator and Purge Diagnostics

OA.3.1 Diagnostics under the own-lag alternatives

The paper reports two alternative specifications that depart from the benchmark in the own-lag treatment alone. On the same validation design, at the same 120-month rolling window, the same cross-state lag order, the same mixing parameter and the same penalty-path floor, the own lags are exempted from the penalty, first at the benchmark lag order and then extended to eight. In every network the buffer around each validation block equals the maximum lag order, two months for the two VAR(2) networks and eight for the eight-own-lag network, so that no training observation shares a lag window with a validation observation. Table OA.2 sets the three networks beside each other, in their raw and factor-adjusted forms, on the quantities that decide whether a decomposition exists at all and how it is composed. These are dynamic stability, the density of the fitted network, the size of the selected penalty, residual dependence in sample and on the validation blocks, and the connectedness index, over all 41 annual networks and over the 21 from 2005, and, at monthly rather than annual frequency, stability and how often penalty selection reaches the floor of the path.

The residual columns summarize first-order dependence rather than a whole autocorrelation profile. The in-sample and held-out entries are medians of the absolute lag-one residual autocorrelation taken across the 48 equations and the annual networks, where the held-out residual of each month is taken from the validation block that did not train on it, at the selected penalty, and the Ljung–Box column is the share of equations rejecting the twelve-lag test at a nominal five percent on those held-out residuals. Freeing the own lags lowers the in-sample median from 0.38 to 0.21 and the held-out median from 0.60 to 0.37, and eight own lags bring them to 0.04 and 0.17, with Ljung–Box rejecting for 37 percent of equations against 100 and 98 percent. Conventional Ljung–Box critical values are not calibrated for equation-specific penalty selection, so these shares describe the residuals rather than test them. By the same token a low rejection count would not establish a serially independent multivariate innovation sequence. What the three networks establish is that the residual dependence of the benchmark can be reduced considerably within the estimator, and what reducing it costs. In the annual networks from 2005, active cross-state slopes per equation fall from 25.4 to 13.2 to 6.7, and in the 16 years from 2010 in which all three are stable mean systemwide connectedness falls from 87.4 to 83.0 to 72.1, as each equation explains more of its own variance before any cross-state term is reached. Over all 41 annual networks the benchmark's in-sample median is 0.39 and its held-out median 0.60, and the own-lags-free and eight-own-lag networks bring the in-sample median to 0.22 and 0.04. Held-out residuals for the raw alternatives are computed only for the annual networks from 2005.

In the factor-adjusted re-estimates, removing each window's national factor lowers systemwide connectedness in every network, from 78.1 to 65.6 at the benchmark, raises the selected penalty, and leaves the eight-own-lag network unstable in three of the 41 annual networks rather than nine. The benchmark's held-out residual dependence is unchanged, 0.60 in both networks.

Table OA.2: Network diagnostics under the own-lag alternatives, raw and factor-adjusted networks

Network	Stable	Max. radius	Cross/ eq.	Own/ eq.	Rel. λ	|ACF(1)| in	|ACF(1)| out	LB out (%)	DYCI	Floor eq.
Panel A. All 41 annual networks, 1985–2025										
Raw networks										
Benchmark	41/41	0.978	23.5	1.62	0.029	0.387	0.602	99.9	78.1	2
Own-lags-free	41/41	0.986	11.2	2.00	0.327	0.218	–	–	71.6	0
Eight-own-lag	32/41	1.074	5.7	8.00	0.628	0.041	–	–	66.0	0
Factor-adjusted networks										
Benchmark	41/41	0.945	21.7	1.67	0.067	0.395	0.598	100.0	65.6	0
Own-lags-free	41/41	0.953	8.1	2.00	0.521	0.221	0.372	97.5	56.2	0
Eight-own-lag	38/41	1.032	4.1	8.00	0.830	0.027	0.167	47.7	48.9	0
Panel B. The 21 annual networks 2005–2025, where the allocation regressions sit										
Raw networks										
Benchmark	21/21	0.978	25.4	1.65	0.018	0.381	0.597	100.0	85.3	1
Own-lags-free	21/21	0.986	13.2	2.00	0.272	0.207	0.370	98.1	80.5	0
Eight-own-lag	16/21	1.036	6.7	8.00	0.475	0.045	0.171	36.6	72.1	0
Factor-adjusted networks										
Benchmark	21/21	0.945	22.0	1.68	0.067	0.385	0.586	100.0	64.8	0
Own-lags-free	21/21	0.909	8.8	2.00	0.475	0.214	0.370	99.6	55.1	0
Eight-own-lag	20/21	1.011	4.2	8.00	0.756	0.028	0.173	44.6	43.4	0
Panel C. Monthly windows, July 1985–December 2025										
Raw networks										
Benchmark	stable in 480 of 486 monthly windows; at least one equation at the penalty floor in 22									
Own-lags-free	stable in 474 of 486 monthly windows; at least one equation at the penalty floor in 0									
Eight-own-lag	stable in 374 of 484 monthly windows; at least one equation at the penalty floor in 0									
Factor-adjusted network										
Benchmark	stable in 484 of 486 monthly windows; at least one equation at the penalty floor in 1									

Note: In Panels A and B the raw rows use the raw networks and the factor-adjusted rows the factor-adjusted re-estimates of the same three specifications. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Networks are estimated at the benchmark specification of Section 2; the own-lags-free network exempts the own-lag coefficients from the penalty, and the eight-own-lag network additionally extends each equation's own lags to eight. The columns report the annual networks that are dynamically stable (companion spectral radius below one), the largest radius, the mean active cross-state and own slopes per equation, the median selected penalty relative to the path maximum, the median absolute lag-one residual autocorrelation across equations and annual networks, in sample and on the validation block that did not train on each month, the share of equations rejecting a twelve-lag Ljung–Box test at 5 percent on those held-out residuals, systemwide connectedness (DYCI, percentage points) averaged over the stable annual networks, and the number of annual-network equations whose selected penalty sits at the path floor. The eight-own-lag network is unstable in the annual networks of 1985, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009 (raw) and 1985, 2003, 2006 (factor-adjusted); the benchmark and the own-lags-free network are stable at all 41 in both. Held-out residuals were computed for the raw own-lag alternatives only in the 21 annual networks from 2005, so their Panel A held-out entries are not available; the factor-adjusted runs kept them in all 41. In the monthly windows only the benchmark was estimated in factor-adjusted form, so Panel C has one factor-adjusted row. Ljung–Box critical values are not calibrated for equation-specific penalty selection, so the rejection shares describe the residuals rather than test them.

OA.3.2 The distance gradient in the raw price data

Whether the flattening of the distance gradient in Figure 3 is a fact about house prices or an artifact of how the network is estimated can be checked without the network. Figure OA.2 computes, in each monthly window and over the same 120 months, the Pearson correlation of monthly log price growth for all 1,128 unordered state pairs, reports how strongly those correlations line up with proximity, and sets that measure beside the estimated gradient of the raw state network, each standardized over the months both cover. No VAR, no penalty and no variance decomposition enters the correlation-based measure, and the two correlate at 0.68. They share their main peak, 0.54 for the raw measure in November 1995 and 0.54 for the estimated gradient in January 1994, and their trough in the mid-2010s, when the correlation-based measure is at or slightly below zero, with a minimum of -0.016 in November 2016, and the estimated gradient falls to 0.06. They diverge in between and at the end. The correlation-based measure falls to 0.14 in September 2005, when the estimated gradient is near a second peak of 0.45, and it recovers to 0.34 in September 2023 before ending the sample at 0.21, while the estimated gradient stays near 0.12. The peak and the trough are therefore properties of the price data, and the estimated gradient's plateau through the mid-2000s and its flatness after 2020 are not reproduced by the correlation-based measure. Comovement itself rose over the same years, the median pair correlation going from 0.19 at December 1985 to 0.73 in the window ending in 2025, and it is in the late, highly synchronized windows that distance explains least about which states move together, until the correlation-based measure recovers after 2020.

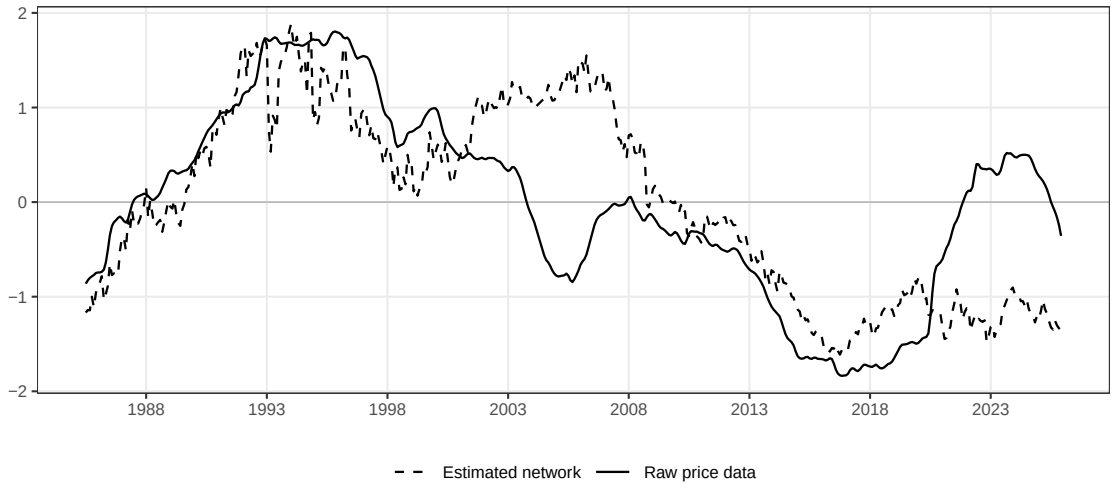


Figure OA.2: Spatial organization of comovement in the raw price data and the estimated distance gradient, January 1985–December 2025

Note: In each monthly window the Pearson correlation of monthly log house-price growth is computed over the trailing 120 months for all $48 \times 47/2 = 1{,}128$ unordered state pairs, without the estimated network. The solid line is the correlation between those 1,128 pair correlations and minus log great-circle distance, the raw-data counterpart of the elasticity in Figure 3, and the dashed line is that elasticity for the raw state network, each standardized over the months in which both exist. The windows are 120 months long and overlap, so the series are smooth by construction and successive values are not independent. The figure is descriptive and no test is reported on it.

OA.3.3 What the purge leaves

Two questions can be asked of the purge. The first is whether the rolling first component is the same factor from one window to the next. One month apart its loadings correlate almost perfectly at the median window and by at least 0.83 at every one. Twelve months apart the median correlation is 0.95, but at a few windows with a weak common factor the component changes identity from one year to the next, so in those networks the purge removes a different common direction. The second is what the purge leaves. The first component's share of standardized growth variance falls from a median of 0.33 to 0.21 across the monthly windows, and the number of factors selected by the Bai and Ng (2002) criterion from about four to two. What remains is regional. After the purge the mean pairwise correlation of states in the same Census region is 0.11 at the median window, against -0.04 for states in different regions. A single-factor purge therefore leaves regional common structure in place, which is why the text reads a factor-adjusted result as structure beyond the national cycle rather than as spillovers among markets. Table OA.6 re-estimates the state network after removing the number of components the criterion selects on each window's raw growth, after which it selects none at 38 of the 41 windows. The sending-side gradient does not survive that purge (Section 6.1). A national-plus-regional de-factoring that separates the two is not run here.

Table OA.3 reports the regression behind the tracking statement of Section 5.2, of a state's share of growth variance on the first principal component on standardized WRLURI and the regressors of Table 2, with year effects and the inference of that table. A standard deviation of WRLURI is associated with 8.07 percentage points more of a state's growth variance on the first component, with a cluster-robust standard error of 1.18 and a 95-percent interval of $[5.42, 10.80]$, against a mean share of 45.9 percent, and lagged log population also enters positively. Panel B describes the share by decade of the window and by quarter of the regulation distribution. The regression is a diagnostic of why the raw receiving-side gradient arises, not a separate finding. It needs no network, only the principal components of each window.

Table OA.3: Regulation and comovement with the national factor, 1985–2025

Panel A. Regression of the share of a state's growth variance on the first component (pp)

	Estimate	Standard error	95-percent interval
WRLURI 2006	8.07***	(1.18)	[5.42, 10.80]
Steep land share	0.61	(1.09)	
log average distance	−2.31	(5.74)	
Lagged log population	2.30**	(0.93)	
Year fixed effects	✓		
State-years	1,968		
Within R^2	0.164		
Mean (SD) of the dependent variable	45.88 (28.84)		

Panel B. Mean share of growth variance on the first component (pp)

	All 48 states	Most regulated quarter	Least regulated quarter
Annual networks 1985–1989	21.48	18.41	21.11
Annual networks 1990–1999	28.01	40.65	17.10
Annual networks 2000–2009	40.51	60.63	21.45
Annual networks 2010–2019	66.68	73.75	55.52
Annual networks 2020–2025	70.26	73.22	65.74
All annual networks 1985–2025	45.88	55.65	35.14

Note: 48 states, 41 annual networks 1985–2025. The dependent variable is 100 times the R^2 of a state's monthly growth on the first principal component of the standardized 48-state growth panel in the network's window, the share of its growth variance that the purge removes; it is computed from raw growth and needs no network. Panel A regresses it on the regressors of Table 2 with year effects; coefficients are percentage points per standard deviation of a standardized regressor and per unit of a logged control, with the 95-percent interval that inverts the bootstrap test in brackets for WRLURI. Panel B averages the share over state-years by period and within the most and least regulated quarters of the WRLURI 2006 distribution (12 states each). Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.3.4 The metropolitan network: estimation and diagnostics

The metropolitan network uses the estimator of Section 2—a rolling elastic-net VAR whose moving-average representation and generalized forecast-error variance decomposition at horizon 12 give the row-normalized connectedness matrix—at the same 41 annual networks and 120-month window. The metropolitan specification chooses the penalty at the minimum of hv-block cross-validation with a buffer around each validation block, on a path floored at 10^{-4}, the design the state benchmark uses. The two scales share the lag order, window, mixing weight, penalty rule, path floor and fold design, so a difference between them is a difference of node set rather than of estimator. With two lags of all 100 series the design has 200 regressors on 120 observations in every equation, so the elastic net is what makes the system computable and the cross-area block is heavily shrunk. Table OA.4 reports the diagnostics that matter for that reading. All 41 annual networks are dynamically stable, and in the 21 from 2005, the years for which the equation diagnostics are computed, the median absolute *in-sample* residual lag-one autocorrelation of 0.33 is below the 0.38 the state network records for the same in-sample object under the same validation design, about 27 cross-area slopes per equation survive shrinkage against 25.4 cross-state slopes at state scale, and one equation of the 2,100 sits at the penalty-path floor. Both autocorrelation figures are in-sample fitted-residual

statistics. Neither is comparable with the median absolute *held-out* lag-one residual autocorrelation of the state network, 0.60 (Table OA.2), which is assembled from the validation block that did not train on the row and is larger for that reason. Averaged over the 21 annual networks from 2005, systemwide connectedness is 93.9 percent, higher than the 85.3 percent of the state network on the same annual basis, because a row-normalized decomposition assigns more of each node's variance to the other 99 nodes than to the other 47. Levels are therefore only partly comparable across system sizes. On the monthly basis the two indexes average 91.2 and 78.2. The windows from 2005 sit higher at both scales. The same table reports the factor-adjusted metropolitan network. It is stable in all 41 annual networks and keeps about 23 cross-area slopes per equation; its systemwide connectedness averages 86.1 percent against 91.2 raw, and the cross-sectional standard deviation of external connectedness over 2005–2025 more than doubles, from 3.1 to 7.0 points.

Section OA.3.1 above reports the state network estimated without shrinkage on the own lags as a robustness check on the benchmark. That check does not survive the move to 100 nodes, and the reason is worth recording. Re-estimated on the same 486 monthly windows with the own-lag penalty switched off, the metropolitan system has a companion eigenvalue outside the unit circle at 188 of them against 4 for the benchmark, and in 16 of the 41 annual networks, 10 of the 21 from 2005, against none. Its median companion radius rises from 0.938 to 0.977 and its maximum from 1.058 to 1.184, and the number of cross-area slopes the penalty retains falls from 26.4 to 12.4 per equation. The explosive windows are concentrated in 2001–2011 and 2021–2025, so the monthly index is not interpretable over exactly the stretches the paper reads. With 200 regressors on 120 observations in every equation, shrinking the own lags along with the rest is what keeps the metropolitan system estimable, and the unshrunk network is reported here as a diagnostic rather than carried into the second stage.

Table OA.4: Metropolitan network diagnostics, raw and factor-adjusted annual networks

	Raw network		Factor-adjusted network	
	1985–2025	2005–2025	1985–2025	2005–2025
Stable annual networks	41 of 41	21 of 21	41 of 41	21 of 21
Maximum companion-matrix radius	0.973	0.970	0.951	0.951
Mean systemwide connectedness (DYCI, pp)	91.2	93.9	86.1	85.6
Mean active cross-metro slopes per equation (of 198)	26.4	27.4	23.2	23.0
Cross-sectional s.d. of external connectedness (pp)	4.6	3.1	6.9	7.0
Median absolute in-sample residual lag-1 autocorrelation	–	0.328	0.343	0.342
Share of equations rejecting Ljung–Box(12) at 5 percent	–	0.98	0.95	0.96
Mean active own slopes per equation (of 2)	–	1.28	1.28	1.29
Equations at the penalty-path floor	–	1 of 2,100	0 of 4,100	0 of 2,100

Note: Columns 1–2 use the raw 100-metro benchmark network and columns 3–4 the factor-adjusted one. The factor-adjusted network is re-estimated after each 120-month window's first principal component of monthly growth is removed (Section 2.3). Entries average over the 41 annual networks 1985–2025 and the 21 from 2005 of the 100-metro elastic-net VAR with 2 cross lags, 2 own lags, own-penalty factor 1, $\alpha = 0.5$, 120 response months, penalty chosen at the minimum of purged hv-block validation (2-month purge), penalty-path floor 10^{-4}, forecast horizon 12, row-normalized GFEVD. The design has 200 regressors on 120 observations in every equation. The cross-sectional s.d. row averages over the annual networks the standard deviation of external connectedness (FROM) across the 100 areas, computed from the networks themselves. The equation-level residual and penalty diagnostics of the raw network were retained only in the annual networks from 2005, so its 1985–2025 column leaves the last four rows blank, while the factor-adjusted run kept them in all 41. The autocorrelation row averages the medians of the annual networks; its $1.96/\sqrt{120}$ band is 0.179. It is an in-sample fitted-residual statistic, not the held-out quantity reported for the state network in Table OA.2.

OA.4 Directional Regressions Across States

OA.4.1 Specification of the directional regressions

Table OA.5 builds the regressions of Table 2 one control at a time, with every coefficient and standard error, for factor-adjusted log TO (Panel A) and log NET (Panel B) and for raw log FROM (Panel C), and reports the WRLURI coefficient with the outcomes in percentage points (Panel D). Column (5) replaces lagged population, which falls inside each price window and may respond to regulation, by the state's 1980 population from the same BEA series. The 1980 value precedes every endpoint, although it lies inside the windows ending in 1985–1989. The factor-adjusted TO association is 7.97 (2.36) without any population control, 7.50 (2.32) with lagged population and 7.55 (2.35) with 1980 population, and log NET is 6.86, 5.83 and 5.76. For raw log FROM the regulation estimate is 2.97 percent with regulation alone, 2.89 once the steep land share enters, 2.70 with average distance, 2.76 with lagged population, the headline, and 2.77 with 1980 population.

Migration openness exists only from 2005, and entering it would confine the directional regressions to half their span. On the 2005–2025 endpoints, with the state means of log migration openness and lagged log GDP in place of population, the raw-network regulation coefficient is 14.3 percent (standard error 5.8) for log TO, 3.0 (0.6) for log FROM and 11.3 (5.8) for log NET, against 10.7, 3.1 and 7.7 with population on the same endpoints.

In levels (Panel D), regulation is associated with 2.12 (0.41) points more external connectedness in the raw network, against a mean of 78.06, so the log and level estimates agree once scaled, with 8.0 (2.9) points more TO connectedness and with a net position 5.9 (2.8) points higher, whose interval, unlike that of log NET, just includes zero. In the factor-adjusted network the three are 4.9 (1.6) points of TO, 1.1 (0.6) of external connectedness and 3.8 (1.5) of NET.

Table OA.5: The directional regressions with the controls entered one at a time, and in levels, 1985–2025

	Controls entered one at a time				1980 population
	(1)	(2)	(3)	(4)	(5)
Panel A. Factor-adjusted network, log *TO*					
WRLURI 2006	9.59***	9.52***	7.97***	7.50***	7.55***
	(2.11)	(2.11)	(2.36)	(2.32)	(2.35)
Steep land share		0.90	−1.02	−0.71	−0.87
		(2.39)	(2.52)	(2.39)	(2.44)
log average distance			16.92	18.69	19.85
			(13.02)	(12.94)	(13.12)
Lagged log population				3.29	
				(2.29)	
log population in 1980					2.53
					(2.43)
Within R^2	0.060	0.060	0.065	0.072	0.069
Panel B. Factor-adjusted network, log *NET*					
WRLURI 2006	6.98***	7.05***	6.86***	5.83**	5.76**
	(2.18)	(2.20)	(2.38)	(2.08)	(2.12)
Steep land share		−0.95	−1.19	−0.51	−0.79
		(2.34)	(2.77)	(2.28)	(2.39)
log average distance			2.13	6.00	9.70
			(13.44)	(12.06)	(12.71)
Lagged log population				7.18**	
				(2.62)	
log population in 1980					6.52**
					(2.76)
Within R^2	0.026	0.027	0.027	0.053	0.047
Panel C. Raw network, log *FROM*					
WRLURI 2006	2.97***	2.89***	2.70***	2.76***	2.77***
	(0.45)	(0.44)	(0.64)	(0.60)	(0.60)
Steep land share		0.98**	0.75	0.72	0.73
		(0.52)	(0.45)	(0.46)	(0.46)
log average distance			2.03	1.82	1.57
			(3.58)	(3.43)	(3.41)
Lagged log population				−0.38	
				(0.67)	
log population in 1980					−0.40
					(0.69)
Within R^2	0.049	0.054	0.055	0.056	0.056
Year fixed effects	✓	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968	1,968

Note: 48 states, 41 annual networks 1985–2025, benchmark network; Panels A and B factor-adjusted, Panel C raw. Column (4) is Table 2, column (4) in Panel A, (6) in Panel B and (2) in Panel C. Lagged population is BEA resident population in the year before the endpoint, inside the endpoint's price window; column (5) replaces it with log population in 1980, which precedes every endpoint. $\log \mathrm{NET} = \log(\mathrm{TO}/\mathrm{FROM})$. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

Table OA.5 (continued). Panel D, outcomes in percentage points

Panel D. Outcomes in percentage points, coefficient on WRLURI 2006	TO	FROM	NET
Raw network	8.011**	2.123***	5.889*
	(2.907)	(0.409)	(2.779)
Factor-adjusted network	4.889***	1.067	3.822**
	(1.613)	(0.649)	(1.523)

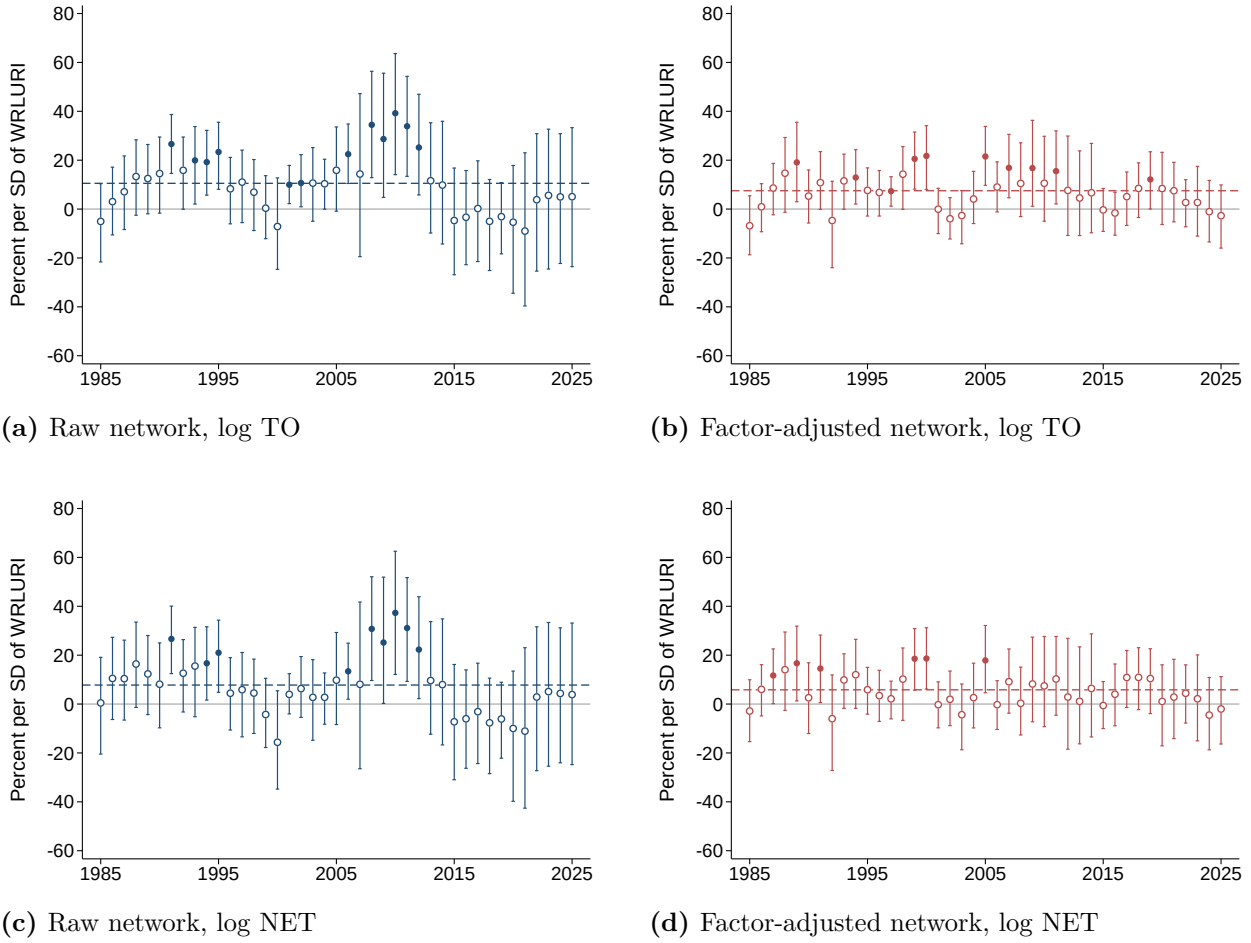
Note: Panel D reports the WRLURI 2006 coefficient of the regressions of Table 2 with the outcomes in percentage points, by least squares with the same regressors and year effects; NET is TO minus FROM. Cluster-robust standard errors by state in parentheses; stars follow the wild cluster bootstrap-t p-values of that table.

OA.4.2 The regulation coefficient endpoint by endpoint

Figure OA.3 estimates the sending-side regressions as a separate 48-state cross-section in every annual network, in both networks. In the factor-adjusted network the TO coefficient is positive in 32 of the 41 years, against 33 raw, and its interval lies above zero in 10, against 12 raw. In no year does the interval lie below zero, in either network or for either sending-side outcome. Averaged by decade, the factor-adjusted coefficient is 7.3 log points over 1985–1989, 9.2 in the 1990s, 9.4 in the 2000s, 6.9 in the 2010s and 2.9 since 2020, and the raw one is about 15 in the 1990s and 2000s and 0.9 since 2020. The sending-side association is therefore present from the first windows, largest in the raw network in the windows that contain the boom and bust of the 2000s, and weakest in both networks in the most recent windows. The common component accounts for 69 percent of a state's growth variance in the windows since 2020, against 22 percent in those ending in 1985–1989, and a smaller residual variance leaves less room for differences between states. Log NET follows the same profile less precisely. It is positive in 33 factor-adjusted years, with an interval above zero in six, about what a constant gradient would produce given the precision of a single cross-section: with standard errors near 6, a gradient of 5.8 in every year would have an interval above zero in about 7 of the 41 years, and for TO, at 7.5, in about 11.

The receiving side has its own profile. In the raw network its coefficient is negative in each year from 1985 through 1988 and largest in the windows that contain the boom and bust of the 2000s. In the windows of the late 1980s, when the common component accounted for only a fifth of a state's growth variance, regulated states took less of their forecast-error variance from elsewhere. In the factor-adjusted network the coefficient averages 1.7 log points against 2.8 raw and correlates 0.49 with the raw profile. The raw coefficient averages −4.8 log points over 1985–1988 and 6.3 log points over 2000–2009, and the factor-adjusted one is positive in 31 of the 41 years, against 36 raw.

The early negative receiving-side coefficients are not the extreme monthly movements of 1981–1983 (Appendix A). Re-estimating the eight annual networks whose windows contain them, 1985–1992, on growth rates winsorized at each state's 1st and 99th percentiles leaves the raw receiving-side coefficient negative in each of 1985 through 1988.



(a) Raw network, log TO

(b) Factor-adjusted network, log TO

(c) Raw network, log NET

(d) Factor-adjusted network, log NET

Figure OA.3: The sending-side regulation coefficient year by year, 1985–2025

Note: Each point is 100 times the WRLURI 2006 coefficient, approximately percent per standard deviation of WRLURI, from a separate 48-state cross-section regression of log TO (a, b) or log NET = log(TO/FROM) (c, d) on the regressors of Table 2 in one annual network, in the raw (a, c) and the factor-adjusted (b, d) benchmark networks. Bars are 95-percent intervals that invert a wild bootstrap-t test imposing the null (9,999 draws, with each state its own cluster in a cross-section). Filled points are years whose interval excludes zero, empty points years whose interval covers it. The dashed line in each panel is the pooled estimate of Table 2, and the four panels share one vertical scale.

OA.4.3 Removing more of the common component

The factor-adjusted network removes one component at the same month, and regional common structure survives it (Appendix C). Table OA.6 re-estimates the state benchmark network under four further purges, each applied within every 120-month window and inside every validation fold. Columns (2) and (3) regress each market on the first component's score at the same month and at lags one to three or one to six, which is the dynamic one-factor representation; column (4) removes two components at every window; and column (5) removes the number of principal components that the IC_{p2} criterion of Bai and Ng (2002) selects on the window's standardized raw growth, with at most eight. That count is chosen once per window and ranges from two to eight, with a median of four. It is highest in the windows ending in 1985–1992 and lowest in those ending in 1993–2004.

After the Bai–Ng purge the criterion selects no further factor at 38 of the 41 windows, systemwide connectedness averages 56.7 percent against 65.6 after the one-component purge, and every annual network is stable.

The sending side survives the lagged scores and two components and not the Bai–Ng count. The regulation gradient on log TO is 5.19 (2.12) with the score at lags one to three, with a wild bootstrap p-value of 0.033 and a 95-percent interval of $[0.50, 9.67]$, and 8.58 (2.20) with lags one to six, against 7.50 at the same month; the lag purges remove 50 and 52 percent of standardized growth variance against 46 for the same-month purge, so the lagged part of the national factor is small and does not carry the gradient. With two components removed the gradient is 5.36 (2.06), with an interval of $[0.92, 9.89]$. Under the Bai–Ng purge it falls to 1.41 (1.14), with a p-value of 0.255 and an interval of $[-1.34, 3.69]$, and the one on log NET to 1.91 (0.96). The within-year cross-state standard deviation of log TO falls from 0.39 after the one-component purge to 0.25 after the Bai–Ng purge, so the stronger purge also compresses the outcome. The in-sample common component of the generalized dynamic factor model, estimated with the fnets package of Barigozzi et al. (2024), was also tried as a purge; at a 120-month window it removes no common variation from these series, so it is not reported.

Table OA.6: Regulation and directional connectedness under five purges of the common component, 1985–2025

	One component and its lagged scores			More components	
	Same month	Lags 1–3	Lags 1–6	Two	Bai–Ng
	(1)	(2)	(3)	(4)	(5)
Panel A. log *TO*					
WRLURI 2006	7.50***	5.19**	8.58***	5.36**	1.41
	(2.32)	(2.12)	(2.20)	(2.06)	(1.14)
Steep land share	−0.71	0.90	−1.49	−1.97	−0.08
	(2.39)	(2.38)	(1.99)	(1.50)	(1.16)
log average distance	18.68	40.35***	33.17**	17.39	6.72
	(12.94)	(12.05)	(12.58)	(11.14)	(6.83)
Lagged log population	3.29	−1.09	−1.77	3.86**	2.15
	(2.29)	(1.89)	(1.98)	(1.77)	(1.31)
Within R^2	0.072	0.109	0.108	0.060	0.017
Panel B. log *FROM*					
WRLURI 2006	1.67	1.08	1.25	1.28	−0.50
	(1.05)	(0.88)	(0.85)	(0.79)	(0.62)
Steep land share	−0.19	−0.15	−1.07	−1.40	−1.14
	(0.87)	(1.12)	(0.82)	(0.99)	(1.34)
log average distance	12.70**	14.53***	17.15***	8.94*	6.84
	(5.32)	(5.27)	(4.63)	(5.10)	(5.58)
Lagged log population	−3.89***	−3.35***	−2.35***	−4.00***	−5.03***
	(0.89)	(0.76)	(0.67)	(0.83)	(0.68)
Within R^2	0.083	0.079	0.071	0.058	0.082
Panel C. log *NET*					
WRLURI 2006	5.83**	4.11**	7.34***	4.08**	1.91*
	(2.09)	(1.59)	(1.76)	(1.71)	(0.96)
Steep land share	−0.51	1.05	−0.42	−0.56	1.06
	(2.28)	(1.60)	(1.64)	(1.75)	(0.89)
log average distance	5.98	25.82***	16.02	8.45	−0.12
	(12.06)	(9.20)	(10.33)	(10.58)	(5.24)
Lagged log population	7.18**	2.26	0.59	7.85***	7.18***
	(2.62)	(1.86)	(2.02)	(2.23)	(1.46)
Within R^2	0.053	0.059	0.064	0.058	0.064
Components removed	1	1	1	2	2–8
Score lags removed	0	1–3	1–6	0	0
Share of growth variance removed	0.459	0.500	0.522	0.580	0.711
Mean DYCI (percent)	65.6	66.7	67.2	61.7	56.7
Year fixed effects	✓	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968	1,968

Note: 48 states, 41 annual networks 1985–2025. Each column is the state benchmark of Table 2, columns (4)–(6), after a purge of the common component applied within each 120-month window and inside each validation fold. Column (1) removes the first principal component of standardized monthly growth at the same month, the paper's factor-adjusted network; columns (2) and (3) also remove its score at lags one to three or one to six; column (4) removes two components, and column (5) the number the Bai and Ng (2002) IC_{p2} criterion selects on the window's raw growth, two to eight with a median of four. Entries are 100 times the coefficients. The share of growth variance removed is the mean over windows and states of the purge regression's R^2, and the mean DYCI is over the 41 annual networks. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.4.4 Amplitude and tracking decomposition

Table OA.7 adds two variables to the regressions of Table 2, in the raw and the factor-adjusted networks, each measured on the same 120-month window as the network. They are the share of a state's growth variance taken by the first principal component of the standardized 48-series panel, and the log standard deviation of its VAR innovation. Both are built from the price series whose forecast-error variance decomposition defines the outcome, and both are plausibly consequences of regulation, so the table is a decomposition of the regulation associations and not a specification of them. It asks how much of each association runs through the size and the national comovement of a state's price surprises.

In the raw network both variables matter. A 1 percent larger innovation standard deviation is associated with about 0.74 percent more TO connectedness and 0.17 percent less external connectedness, and a state whose growth loads more heavily on the first principal component has more external connectedness, 0.27 percent per percentage point of explained variance. Holding both fixed cuts the external-connectedness association from 2.76 to 1.12 percent (0.68), the TO association from 10.5 to 5.3 (3.5) and log NET from 7.8 to 4.2 (3.4), none of them then distinguishable from zero. In the factor-adjusted network the TO association is 7.5 percent (2.3) without them and 6.8 (2.3) with them, so the sending side beyond the national cycle does not run through the size of a state's own surprises. Factor-adjusted log NET falls from 5.8 (2.1) to 3.1 (1.8). The factor-adjusted external-connectedness association rises from 1.7 (1.0) to 3.6 (1.1) once the two variables enter. That rise should not be read as a receiving-side result, because in the factor-adjusted network the first-component share measures how much of each state's variation the purge removed, so conditioning on it conditions on the purge itself.

Table OA.7: Directional connectedness with the amplitude controls added, raw and factor-adjusted networks, 1985–2025

	Raw			Factor-adjusted		
	log TO	log FROM	log NET	log TO	log FROM	log NET
	(1)	(2)	(3)	(4)	(5)	(6)
WRLURI 2006	5.35	1.12	4.23	6.77***	3.65***	3.13*
	(3.54)	(0.68)	(3.38)	(2.30)	(1.12)	(1.79)
First-component variance share (pp)	0.36***	0.27***	0.09	−0.02	−0.18***	0.17**
	(0.11)	(0.02)	(0.11)	(0.08)	(0.03)	(0.07)
log innovation SD	73.56***	−17.12***	90.68***	48.58***	−27.81***	76.38***
	(6.55)	(1.43)	(7.01)	(4.82)	(2.29)	(5.36)
Steep land share	0.43	0.68	−0.25	−1.28	0.25	−1.53
	(2.96)	(0.54)	(2.79)	(2.72)	(0.96)	(2.86)
log average distance	−5.34	8.33**	−13.67	−0.15	23.04***	−23.19
	(14.95)	(3.24)	(14.06)	(12.94)	(5.74)	(12.63)
Lagged log population	−4.84	2.74***	−7.59*	−6.72**	2.29**	−9.00***
	(3.87)	(0.69)	(3.92)	(2.05)	(0.91)	(1.93)
Year fixed effects	✓	✓	✓	✓	✓	✓
Within R^2	0.318	0.413	0.358	0.243	0.336	0.398

Note: 48 states, 41 annual networks 1985–2025; raw network in columns (1)–(3), factor-adjusted in (4)–(6). The regressions of Table 2 with two amplitude controls added, each measured on the network's own window: the state's share of growth variance on the first principal component of the standardized 48-series panel, in percentage points, computed from raw growth and so the same in both networks, and the log standard deviation of the state's VAR innovation in the network. Both are functions of the price series that define the outcome, so the table decomposes the headline associations rather than specifying them, and in the factor-adjusted network the first-component share is what the purge removed. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.5 The Labor-Demand Interaction and the Demand Proxies

Table 3 of the paper reports the interaction of regulation and the labor-demand shock on all three sides of a metropolitan area's position. Table OA.9 reports the receiving side, log FROM, in four specifications on the factor-adjusted network: regulation alone (column 1), the steep land share alone (column 2), both, which is the specification of the paper's table (column 3), and both split at the year's average shock (column 4). The checks below are run on column (1). Its sending-side version, the same four columns on log TO with the same 100 areas, controls and shock, gives a regulation gradient of TO at the average shock of 2.9 (1.7) that rises with the shock by 1.9 (0.9), with a wild bootstrap p-value of 0.05. Beside regulation the steep land share interaction is 0.6 (1.1). Split at the average shock, the regulation interaction is 2.6 (1.7) above the average and 0.8 (1.9) below it, and neither their difference, 1.8 (2.9), nor their sum, 3.4 (2.1), is distinguishable from zero.

Table OA.8 sets the checks of column (1) of Table OA.9 side by side for the two sides of an area's position: inference clustered by dominant industry, each Census division and the five highest-demand areas left out, the industry-classification eras, and the other margins. On the receiving side every check of column (1) leaves the interaction between 0.63 and 0.96, with p-values below 0.02, and it holds within areas over time. On the sending side the estimates stay between 1.4 and 2.5 when areas or endpoints are left out, but ten of those thirteen checks have p-values above 0.05, and within areas over time the interaction is 0.8 (1.1). Neither side shows the interaction in the raw network, and across states both are imprecise.

Table OA.8: Checks of the labor-demand interaction on the receiving and the sending side, metropolitan areas and states, 1985–2025

	log FROM		log TO		
	WRLURI × S (SE)	p	WRLURI × S (SE)	p	Clusters / obs.
Panel A. Inference and specification					
Table 3, column (1)	0.76*** (0.21)	0.001	1.86* (0.85)	0.050	100 / 4,100
Clusters by dominant industry in 1975	0.76*** (0.16)	0.010	1.86* (0.69)	0.055	16 / 4,100
Column (4), WRLURI × S^+, industry clusters	1.24** (0.55)	0.037	2.59 (1.95)	0.171	16 / 4,100
Full natural-amenity scale and its interaction added	0.85*** (0.24)	0.003	2.11* (1.02)	0.068	100 / 4,100
Panel B. Areas left out					
Without East North Central	0.70*** (0.23)	0.007	1.87* (0.89)	0.065	87 / 3,567
Without East South Central	0.71*** (0.22)	0.003	1.77** (0.80)	0.044	93 / 3,813
Without Middle Atlantic	0.74*** (0.22)	0.003	1.47 (0.88)	0.117	88 / 3,608
Without Mountain	0.63*** (0.21)	0.005	2.15** (0.83)	0.029	92 / 3,772
Without New England	0.86*** (0.22)	0.000	2.31** (0.89)	0.027	92 / 3,772
Without Pacific	0.83*** (0.23)	0.002	1.80 (1.04)	0.106	86 / 3,526
Without South Atlantic	0.76*** (0.23)	0.003	1.38 (1.06)	0.249	79 / 3,239
Without West North Central	0.94*** (0.21)	0.001	1.78* (0.93)	0.077	94 / 3,854
Without West South Central	0.67*** (0.24)	0.005	1.91 (1.02)	0.114	89 / 3,649
Without San Jose, San Francisco, Seattle, Boston, Austin	0.68*** (0.22)	0.005	1.56 (0.95)	0.142	95 / 3,895
Panel C. Endpoints					
SIC endpoints	0.96*** (0.27)	0.003	1.89 (1.65)	0.317	100 / 1,600
NAICS endpoints	0.74** (0.23)	0.013	2.46 (1.31)	0.136	100 / 1,500
Endpoints outside the chained windows	0.86*** (0.17)	0.000	1.64* (0.86)	0.072	100 / 3,100
Panel D. Other margins					
Area effects, factor-adjusted	0.69* (0.28)	0.055	0.84 (1.09)	0.472	100 / 4,100
Area effects, raw	0.04 (0.21)	0.853	−1.28 (1.22)	0.289	100 / 4,100
Raw network	0.01 (0.16)	0.955	−0.87 (1.09)	0.428	100 / 4,100
48 states, factor-adjusted	1.35 (1.04)	0.247	−1.50 (2.36)	0.560	48 / 1,968
48 states, raw	0.86 (0.69)	0.242	−2.07 (3.34)	0.563	48 / 1,968

Note: Each row is the interaction of WRLURI 2006 with the labor-demand shock S in column (1) of Table OA.9 (log FROM) and of its sending-side version (log TO), re-estimated as the row describes, unless the row names column (4); rows without a network are factor-adjusted. Dominant-industry clusters group the areas by the industry with the largest wage-bill share in 1975 (16 clusters). The amenity row adds the full natural-amenity scale and its interaction with the shock to the climate control. Endpoint rows keep the shock of the full sample; the chained windows span the 1997 change of industry classification. Area effects replace the Census-division effects. The state rows use the controls of Table 2 with year effects and state clusters. Entries are 100 times the coefficients, with cluster-robust standard errors in parentheses; p is the wild cluster bootstrap-t p-value that imposes the null (9,999 draws), and stars follow it. The last column gives clusters and observations.

Table OA.9: Regulation, terrain and relative labor demand on the metropolitan receiving side, four specifications, factor-adjusted network, 1985–2025

Dependent variable	log FROM			
	(1)	(2)	(3)	(4)
WRLURI 2006	0.51		0.53	0.11
	(0.37)		(0.38)	(0.50)
WRLURI × S	0.76***		0.78***	
	(0.21)		(0.22)	
WRLURI × S^+				1.24**
				(0.37)
WRLURI × S^-				0.19
				(0.41)
Steep land share		0.37	0.45	1.03**
		(0.42)	(0.39)	(0.44)
Steep land share × S		0.15	−0.04	
		(0.20)	(0.18)	
Steep land share × S^+				−0.64*
				(0.29)
Steep land share × S^-				0.87**
				(0.30)
Labor-demand shock S	−0.19	−0.36	−0.18	
	(0.20)	(0.25)	(0.20)	
S^+				−0.25
				(0.34)
S^-				−0.09
				(0.34)
Controls and same-state sample partners	✓	✓	✓	✓
Climate amenity index	✓	✓	✓	✓
Year and Census-division effects	✓	✓	✓	✓
Areas	100	100	100	100
Area-years	4,100	4,100	4,100	4,100
Within R^2	0.259	0.250	0.260	0.266

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025, factor-adjusted network. The dependent variable is log FROM, the log of an area's external connectedness. The labor-demand shock S is predicted labor demand, the shift-share prediction of an area's earnings growth over the window (Section OA.5.2), standardized across areas in each annual network; $S^+ = \max(S, 0)$ and $S^- = \min(S, 0)$. WRLURI 2006 and the steep land share are standardized across areas. Controls are the climate amenity index, the four climate components of the USDA natural amenity scale, lagged log population, the outside-node-set share and the number of an area's same-state sample partners, with year and Census-division effects. In column (4) the WRLURI slope above the average minus that below it is 1.05 (0.65, $p = 0.158$) and their sum 1.42 (0.43, $p = 0.012$); for the steep land share they are −1.51 (0.46, $p = 0.014$) and 0.22 (0.37, $p = 0.608$). Entries are 100 times the coefficients, approximately the percent difference in FROM per standard deviation of a standardized regressor. Within R^2 is net of the fixed effects. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.5.1 The metropolitan node set

The metropolitan node set is itself a sample choice, and it bears on the economics. Ranking areas by 2000 population puts eleven of the 100 in California, eight in Florida, seven in Ohio and six each in New York, Pennsylvania and Texas, and leaves nine states without an area, while areas in the same state share state shocks that the one-factor purge leaves in place. Table OA.10 re-estimates both networks, with the benchmark specification, on an apportioned node set of 100 areas and on one of 75. The 100-area set gives each of the 48 contiguous states at least one area and divides the rest by the Huntington–Hill method on the square root of each state's metropolitan population, taking the largest areas first within a state. It shares 78 areas with the benchmark, gives no state more than six, and covers 75 percent of the ranking pool's 2000 population against 77 percent. The 75-area set is built by the same rule with 75 seats and gives no state more than four.

The allocation result holds. Among pairs in different states the factor-adjusted exposure coefficient under the geography and node-year design of Table 4 is 0.052 (0.013) on the 100-area set and 0.062 (0.022) on the 75-area set, against 0.060 (0.025), and across all pairs it is 0.201 and 0.186 against 0.171. The receiving-side demand interaction keeps its sign and most of its size but not its precision. Column (1) of Table OA.9 gives 0.60 (0.34) on the 100-area set, estimated on 99 of its areas because the Wharton survey does not cover Cheyenne, Wyoming, and 0.82 (0.42) on the 75-area set, against 0.76 (0.21) on the benchmark, and the slope above the average shock is 0.92 (0.55) and 1.39 (0.58) against 1.24 (0.37). The sending-side interaction does not survive. On the 100-area set it is 0.8 (1.2), against 1.9 (0.9), and the slope above the average shock falls from 2.6 to 0.8 (1.7); on the 75-area set it is 0.59 (1.24). That interaction is specific to the benchmark node set.

Table OA.10: Metropolitan results on the benchmark node set and on apportioned node sets of 100 and 75 areas

	Benchmark, 100 largest		Apportioned, 100		Apportioned, 75	
	Raw (1)	Factor-adjusted (2)	Raw (3)	Factor-adjusted (4)	Raw (5)	Factor-adjusted (6)
Panel A. Table OA.9, log *FROM*						
WRLURI × S, column (1)	0.01	0.76***	−0.00	0.60	−0.03	0.82*
	(0.16)	(0.21)	(0.22)	(0.34)	(0.33)	(0.42)
WRLURI × S^+, column (4)	0.54**	1.24**	0.39	0.92	0.65	1.39*
	(0.26)	(0.37)	(0.30)	(0.55)	(0.47)	(0.58)
WRLURI × S^-, column (4)	−0.63	0.19	−0.20	0.35	−0.58	0.25
	(0.30)	(0.41)	(0.44)	(0.69)	(0.58)	(0.90)
Panel B. The same regressions, log *TO*						
WRLURI × S, column (1)	−0.87	1.86*	−0.96	0.76	−2.09	0.59
	(1.09)	(0.85)	(1.41)	(1.17)	(1.37)	(1.24)
WRLURI × S^+, column (4)	−2.42	2.59	−2.97	0.78	−3.03	1.44
	(2.31)	(1.72)	(2.57)	(1.68)	(2.79)	(1.81)
WRLURI × S^-, column (4)	0.39	0.81	0.38	0.76	−1.39	0.28
	(2.58)	(1.85)	(2.89)	(2.41)	(2.91)	(2.59)
Areas, Panels A and B	100	100	99	99	74	74
Panel C. Table 4, column (4), migration exposure						
All pairs	0.060***	0.171***	0.066***	0.201***	0.056***	0.186***
	(0.022)	(0.056)	(0.018)	(0.046)	(0.019)	(0.053)
Pairs in different states	0.009	0.060**	0.017***	0.052***	0.020**	0.062***
	(0.010)	(0.025)	(0.007)	(0.013)	(0.009)	(0.022)
Pairs in different states, area-year pairs	200,256	200,256	204,204	204,204	115,038	115,038
States with an area	39	39	48	48	48	48
Most areas in one state	11	11	6	6	4	4

Note: Columns (1)–(2) are the benchmark node set, the 100 largest contiguous-US metropolitan areas by 2000 population; columns (3)–(4) an apportioned node set of 100 areas that gives each of the 48 states at least one area and divides the rest by the Huntington–Hill method on the square root of each state's metropolitan population, taking the largest areas first within a state (78 areas in common with the benchmark); columns (5)–(6) the same rule with 75 seats, no state having more than four. Every second-stage variable is rebuilt on the node set's own network, and the supply proxies and the labor-demand shock are standardized within it. Panel A repeats the regressions of Table OA.9, columns (1) and (4), with the same controls, on the areas with a Wharton index value (the survey does not cover Cheyenne, Wyoming), and Panel B the same regressions with log TO; entries are 100 times the coefficients, with standard errors clustered by area and wild cluster bootstrap-t p-values (9,999 draws). Panel C repeats the geography and node-year design of Table OA.14, column (4), on pairs in different states, 2005–2025, with shared-node standard errors and p-values from $t(G-1)$, G the number of areas. *, ** and *** denote $p < 0.10$, 0.05 and 0.01.

OA.5.2 The demand proxies

Table OA.11 adds two measures of demand to the sending-side regressions, labor demand predicted by industry mix and natural amenities, across states (Part 1) and across metropolitan areas (Part 2). Both are built without the price data, and neither enters the networks. Column (4) of Part 2 interacts regulation with the labor-demand shock on the sending side, and Table 3 reports the interaction on all three sides.

Predicted labor demand is a shift-share prediction (Bartik, 1991). For market j and the window ending with year t,

$$B_{jt} = \sum_k s_{jk,t-10} \left[\log E_{k,t}^{(-j)} - \log E_{k,t-10}^{(-j)}\right],$$

where $s_{jk,t-10}$ is industry k's share of the market's wage bill in the year before the window, and $E_{k,t}^{(-j)}$ is national industry-k wages in year t excluding the market itself.

- States use BEA wages and salaries by industry (SAINC7S and SAINC7N). Metropolitan areas use BEA county earnings by industry (CAINC5S and CAINC5N), summed over each CBSA's counties.
- Industries are the finest non-overlapping level, 73 SIC and 87 NAICS cells. Construction, real estate and the military are excluded, because house prices move them directly.
- Suppressed cells, 0.8–1.6 percent of the state wage bill and 4–5 percent of metropolitan earnings, are filled top-down within the published industry tree.
- Windows that span the change from SIC to NAICS are chained, with SIC growth up to the break, NAICS growth after it and NAICS shares at the break. The state tables overlap in 1998–2001, and the state shocks of the 2002–2007 networks are chained at 1998. The metropolitan SIC series ends in 2000 and the NAICS series begins in 2001, so the metropolitan shocks of the 2001–2010 networks use SIC growth to 2000 and NAICS growth from 2001, and the one-year 2000–2001 link is omitted. The shock of the 2025 network ends in 2024. The chained and all-SIC shocks agree closely for the windows ending in 1999 and 2000 (correlation 0.99) but not for 2001, so the six state annual networks whose windows cross the break (2002–2007) are left out of the shock-magnitude column; including them gives 7.23. The state interaction of Section 5.1 includes them.

The labor-demand shock is predicted labor demand relative to the year's average across markets, in units of the within-year standard deviation averaged over the classification era e,

$$S_{jt} = (B_{jt} - \bar{B}_t)/\bar{\sigma}_{e(t)}.$$

Tables 3 and OA.9 use instead the prediction standardized across markets in each annual network, $(B_{jt} - \bar{B}_t)/\mathrm{sd}_t(B)$, with mean zero and standard deviation one in every window. The shock magnitude is $|S_{jt}|$, divided by its standard deviation within the era. It measures how far an area's long-run

relative demand growth, predicted over the ten years of the window, departs from the year's mean, and not the size or the volatility of demand surprises within the window.

The proxy measures labor demand, not an instrument, so no exclusion restriction is involved. It predicts actual growth over the same window in regressions of actual on predicted growth.

- States: slope 1.39 (0.35) on wage-bill growth, $F = 16.2$.
- Metropolitan areas: slope 1.45 (0.26) on earnings growth, $F = 30.9$.

Its magnitude, the shock magnitude defined above, is nearly uncorrelated with the regulation index (-0.12 across states, -0.08 across metropolitan areas). The standardized prediction itself is somewhat higher in regulated markets. Its correlation with the index averages 0.36 across states and 0.20 across metropolitan areas within a window, and ranges from 0.06 to 0.62 and from -0.06 to 0.38.

Natural amenities are the USDA scale of McGranahan (1999), a county index of winter temperature, winter sun, summer temperature, summer humidity, topographic variation and water area. It is aggregated with 1990 population weights and standardized within each sample. It correlates 0.33 with the regulation index across states and 0.45 across metropolitan areas, where the highest-amenity areas are coastal California. It also correlates with the slope share (0.52 across states, 0.61 across metropolitan areas), which shares its topographic component, so the amenity columns replace the steep land share rather than add to it.

Across states, neither measure moves the factor-adjusted sending side (Table OA.11, Part 1). The association is 7.2 percent with the shock magnitude held fixed and 7.9 with amenities held fixed. In the raw network, amenities predict TO strongly, 19.5 percent (6.0), and the regulation association rises to 12.2 (3.1). The amenity component of the sending side therefore runs through the national cycle.

Across metropolitan areas (Part 2), amenities predict factor-adjusted TO, 14.9 percent (4.3), and holding them fixed lowers the regulation coefficient from 4.6 to 3.2 (columns 1 and 3).

Column (4) interacts regulation with the labor-demand shock. The factor-adjusted sending side is steeper where relative labor demand is stronger, by 3.0 percent per standard deviation of each (1.1). That is the ordering a supply reading implies. Part 2 does not hold fixed the number of same-state sample partners, a property of the node set. With it held fixed, and the shock standardized within each window, the interaction is 1.9 (0.9), as above. In the raw network the interaction is -0.8 (1.1) on total TO, and the regulation gradient of raw TO is 6.6 (2.6) (column 6). Table OA.8 checks the receiving-side interaction of Table OA.9, column (1), against industry clusters, amenities, leave-outs and the classification periods. In the specification of Part 2, the comparison is across metropolitan areas. With area effects the interaction is 1.0 (1.1, clustered by area), not distinguishable from zero, and across states it is -1.5 (2.5).

Split at the year's average shock with the number of same-state sample partners held fixed, the sending-side interaction is 2.6 (1.7) above the average and 0.8 (1.9) below it, a split at the

within-year median gives the same pattern, and without the partner control the two are 6.7 and -1.2, a difference of 7.9 (3.2) with an interval of $[1.1, 15.0]$.

Durable housing implies an asymmetry (Glaeser and Gyourko, 2005). A constraint limits new construction when demand rises, but when demand falls the housing stock contracts slowly whatever the regulation. With the partner count held fixed the split does not detect that asymmetry. The shock is also relative. Predicted real growth, deflated by the CPI-U over the same ten years, is negative in 0.6 percent of area-windows, from 9 areas, so the split at the year's average compares above- with below-average growth rather than rising with falling demand.

Table OA.11: Sending side with demand proxies, states and metropolitan areas, factor-adjusted and raw networks, 1985–2025

Part 1. States

Dependent variable:	log TO			
	(1)	(2)	(3)	(4)
Panel A. Factor-adjusted network				
WRLURI 2006	7.50***	7.50***	7.20***	7.93***
	(2.32)	(2.31)	(2.29)	(2.25)
Shock magnitude			−2.27	
			(1.52)	
Natural-amenity scale				2.91
				(4.76)
Steep land share	−0.71	0.24	0.30	
	(2.39)	(2.31)	(2.16)	
log average distance	18.69	19.78	20.06	4.89
	(12.94)	(12.30)	(11.87)	(23.10)
Lagged log population	3.29	2.46	2.39	2.72
	(2.29)	(2.20)	(2.10)	(2.70)
State-years	1,968	1,680	1,680	1,968
Within R^2	0.072	0.074	0.077	0.073
Panel B. Raw network				
WRLURI 2006	10.54**	9.93**	9.20**	12.22***
	(3.62)	(3.92)	(3.81)	(3.08)
Shock magnitude			−5.64	
			(3.43)	
Natural-amenity scale				19.48**
				(6.04)
Steep land share	1.19	2.10	2.26	
	(3.43)	(3.63)	(3.68)	
log average distance	19.12	20.72	21.40	−56.27*
	(16.96)	(17.67)	(17.58)	(27.57)
Lagged log population	12.06**	11.33**	11.14**	7.87
	(3.92)	(4.22)	(4.10)	(4.12)
State-years	1,968	1,680	1,680	1,968
Within R^2	0.097	0.086	0.094	0.127
Year fixed effects	✓	✓	✓	✓
Annual networks	41	35	35	41

Note: Part 1: 48 states, 41 annual networks 1985–2025; Panel A factor-adjusted, Panel B raw. Column (1) is Table 2, column (4) in Panel A and column (1) in Panel B. Column (2) is the same regression on the 35 annual networks outside the windows that cross the SIC–NAICS change (2002–2007), whose shocks are chained, and column (3) adds the shock magnitude, the absolute deviation of predicted ten-year labor-demand growth from the year's average across states in units of its standard deviation within the classification era (Online Appendix Section OA.5.2). Column (4) replaces the steep land share with the natural-amenity scale. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

Table OA.11 (continued). Part 2, metropolitan areas

Dependent variable: log TO	Factor-adjusted network					Raw network
	(1)	(2)	(3)	(4)	(5)	(6)
WRLURI 2006	4.55	4.67	3.23	4.67	3.30	6.58**
	(2.97)	(2.94)	(2.75)	(2.92)	(2.69)	(2.60)
Shock magnitude		0.87				
		(1.28)				
Natural-amenity scale			14.87***		15.10***	
			(4.28)		(4.40)	
Labor-demand shock				−0.55	−1.12	
				(1.23)	(1.23)	
WRLURI × shock				3.02**	2.94*	
				(1.13)	(1.39)	
Natural-amenity scale × shock					−0.49	
					(1.85)	
Steep land share	−1.06	−1.03		−0.97		1.29
	(3.48)	(3.47)		(3.47)		(3.29)
Lagged log population	−4.39	−4.40	−3.26	−4.56	−3.19	0.48
	(2.78)	(2.77)	(2.73)	(2.81)	(2.73)	(2.74)
Outside-node-set share	−13.53***	−13.41***	−8.18*	−13.78***	−8.39**	−13.29***
	(3.28)	(3.24)	(3.81)	(3.22)	(3.72)	(3.49)
Within R^2	0.081	0.081	0.113	0.088	0.120	0.074
Year fixed effects	✓	✓	✓	✓	✓	✓
Census-division fixed effects	✓	✓	✓	✓	✓	✓

Note: Part 2: the 100 largest metropolitan areas, 41 annual networks 1985–2025, 4,100 area-years; columns (1)–(5) factor-adjusted and column (6) raw, repeating column (1). The 95-percent wild bootstrap intervals of the WRLURI coefficient are $[-2.1, 10.6]$ in column (1) and $[1.0, 11.9]$ in column (6). The labor-demand shock S is predicted labor demand relative to the year's average across areas, in units of its within-year standard deviation, and the shock magnitude is $|S|$. Columns (3) and (5) replace the steep land share with the natural-amenity scale. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.6 The Metropolitan Receiving Side and the Destination Split

This appendix documents the metropolitan system that Section 5 places beside the 48-state analysis. It describes how the node set is built and how representative it is, how the network behaves, and how the receiving-side result depends on the geographic controls and on the removal of the national factor. The design variations and the allocation constructions are in the rest of this section and in Section OA.7.

OA.6.1 Node set, covariates and exposure

The metropolitan system is the 100 largest core-based statistical areas of the contiguous United States by Census 2000 population, the counterpart of the 48-state rule. Ranking precedes every screen. A CBSA whose FMHPI series carries a flagged growth month over the estimation span is reported and kept rather than replaced by a smaller market, so the node set is a function of size alone. The price series are the seasonally adjusted Freddie Mac CBSA indexes, transformed exactly as in equation (1). Two coding changes between the 2020 and 2023 CBSA delineations (Cleveland, Poughkeepsie) are reconciled before matching, and metropolitan areas that were split or merged between the 1999 metropolitan definitions used by the Wharton survey and the elasticities of Saiz (2010) and the current CBSAs inherit those measures from every 1999 area of whose 2010 housing stock they hold at least ten percent.

Two checks bear on the top-100 rule. The average of member growth weighted by 2010 housing stock tracks the national FMHPI with a correlation of 0.994 over 1976–2025, the span the networks use, and a terminal index ratio of 0.973. The fit is closer from 1991 (0.998) than over 1976–1990 (0.984), and weights built from the 1980 or 1990 housing counts leave the early correlation at 0.981 and 0.985. The 100 areas hold 63 percent of the 2010 national housing stock. The system is a large-market system, not a national one, and its connectedness measures are read cross-sectionally for that reason. The selection tests show what the rule selects on. The 100 are more regulated than the other contiguous-US CBSAs, by 0.34 index points or 0.43 of the CBSA-level standard deviation of WRLURI 2006 (0.78 across the 275 CBSAs with the measure), and less elastic, and they are no different in terrain. Selection runs in the direction of the regressor of interest and is stated rather than assumed.

The metropolitan WRLURI 2006 is the equal-jurisdiction mean over the responding jurisdictions of the 1999 metropolitan areas mapped to each CBSA, the rule used for the states (Section OA.1.3). The metropolitan values the index authors publish, as carried in the Saiz (2010) file, instead weight jurisdictions by the survey's metropolitan sampling weights. The two constructions correlate at 0.987 across the 256 areas in both files. Terrain is the steep land share over the CBSA polygon, from the same source as the state measure. Saiz's elasticity and unavailable-land share are attached where they exist and, because the elasticity is constructed from regulation and unavailable land, the elasticity is never entered beside either. Each area is assigned a coastal, sunbelt or interior belt. It is coastal if its internal point lies within 100 kilometers of the ocean coastline, otherwise sunbelt

if south of 37°N, and otherwise interior. Migration exposure is built from IRS county-to-county flows aggregated to CBSA pairs with the paper's $t-15$ to $t-11$ pool. Unlike the state matrices, the metropolitan flow matrices hold movers only, so an area's diagonal counts moves between its own counties, its external openness averages about 70 percent, and the common-origin term of equation (8) is 39 percent of external exposure (Table OA.20). Each area is assigned the *principal state* holding the largest share of its 2010 housing units (Washington, D.C., whose largest share lies in Virginia, is the one multistate area not assigned to the first state in its title). The assignment is fixed across years, although it postdates the windows ending before 2010 and falls inside those ending 2010 to 2019, and it is available in all 21 annual networks from 2005. The 100-area system is also open. Only 61 percent of migrating exemptions have both ends inside it, against essentially all for the 48 states, so normalizing flows within the system conditions on staying inside it. Routing the remainder through an absorbing node instead changes the allocation coefficients little (Section OA.7.2).

OA.6.2 External connectedness across fixed-effect structures

Table OA.12 reports the metropolitan receiving-side regressions summarized in Section 5.2, in the raw and the factor-adjusted networks. Without the two population controls, a one-standard-deviation increase in metropolitan regulation is associated with 1.3 percent more external connectedness (0.3) in the raw network, 0.9 (0.3) with them, 0.5 (0.1) within states, and 0.5 (0.1) with principal-state-by-year effects, with a wild-cluster p-value of 0.009 over the twelve principal states. After the purge the four columns give 0.96, 0.71, 0.35 and 0.34, with standard errors roughly twice as large and wild-cluster p-values of 0.31 and 0.33 for the within-state columns; the paired differences from the raw estimates, -0.17 to -0.35, are not distinguishable from zero.

Two features of the metropolitan comparison need stating. The first is that the two scales do not condition on quite the same controls. Both carry lagged log population as the size control, and the division and state effects do at this scale what average distance does across 48 states. Coefficients at the two scales can be placed on a common footing by dividing each by the within-year cross-sectional standard deviation of log external connectedness in its own node set, 0.122 across the 48 states and 0.054 across the 100 metropolitan areas. On that scale, in the raw networks, a standard deviation of regulation moves a state 0.23 standard deviations of the state distribution and a metropolitan area 0.16 with the population controls and 0.10 within states, so the metropolitan gradient is about 0.4 to 0.7 of the state one. The comparison is approximate. The regressor is standardized within each node set, and the top-100 node set is more regulated than the other CBSAs.

The second is a control the state panel does not need. The 48 states cover the country and the 100 metropolitan areas do not, and an area whose neighbors lie mostly outside the node set has less of the estimated system to be connected to. The regressions therefore carry the share of the population within 200 kilometers that lives in counties belonging to no sample area, measured in the calendar year before each network's year. It enters negatively, as that reading implies, at -1.6 percent per standard deviation with division effects and -0.7 within states in the raw network, and entering it with population lowers the regulation association from 1.3 to 0.9 percent under division

effects. Neither conclusion depends on the 200-kilometer radius. Rebuilt at 100 and 300 kilometers, the raw regulation association is 1.11 and 0.97 percent with division effects and 0.55 and 0.48 within states, distinguishable from zero in every case, and the control itself is negative at every radius and weakest at 100 kilometers, where much of what lies around an area's internal point is the area's own counties. The association is 0.89 percent over the 2005–2025 endpoints and 0.88 over 1985–2004 with division effects, and 0.64 and 0.39 within states, the earlier estimates less precise.

Table OA.12: Regulation and external connectedness across metropolitan areas, raw and factor-adjusted networks, 1985–2025

Dependent variable	log external connectedness (FROM)			
	(1)	(2)	(3)	(4)
Panel A. Raw network				
WRLURI 2006	1.31***	0.88**	0.52***	0.51***
	(0.30)	(0.31)	(0.14)	(0.15)
Steep land share	0.24	0.18	−0.23	−0.23
	(0.47)	(0.43)	(0.16)	(0.18)
Lagged log population		−0.05	0.07	0.10
		(0.26)	(0.13)	(0.13)
Outside-node-set share		−1.60***	−0.71***	−0.70**
		(0.31)	(0.23)	(0.27)
Wild-cluster p, principal states	–	–	0.009	0.009
Within R^2	0.033	0.075	0.047	0.112
Panel B. Factor-adjusted network				
WRLURI 2006	0.96	0.71	0.35	0.34
	(0.61)	(0.65)	(0.29)	(0.32)
Steep land share	0.11	0.01	0.06	0.06
	(0.72)	(0.71)	(0.23)	(0.25)
Lagged log population		−0.91	0.24	0.25
		(0.61)	(0.23)	(0.25)
Outside-node-set share		−1.74**	−0.10	−0.16
		(0.57)	(0.23)	(0.26)
Wild-cluster p, principal states	–	–	0.309	0.327
Within R^2	0.008	0.033	0.008	0.019
WRLURI 2006, factor-adjusted minus raw	−0.35	−0.17	−0.17	−0.17
	(0.45)	(0.45)	(0.26)	(0.28)
Year fixed effects	✓	✓	✓	–
Census-division fixed effects	✓	✓	–	–
Principal-state fixed effects	–	–	✓	–
Principal-state-by-year fixed effects	–	–	–	✓
Sample	All	All	States with ≥ 3 metros	States with ≥ 3 metros
Metropolitan areas	100	100	64	64
Metro-years	4,100	4,100	2,624	2,624

Note: The 100 largest metropolitan areas of the contiguous United States, 41 annual networks 1985–2025; Panel A raw, Panel B factor-adjusted. The dependent variable is the log of external connectedness (FROM). An area's principal state is the state holding the largest share of its 2010 housing units; columns 3 and 4 use the 64 areas of the 12 principal states with at least three sample areas, and the principal-state-by-year effects of column 4 absorb each state's shocks in every year. WRLURI and the steep land share are standardized within the node set, population is the CBSA's county population in the year before the network's year, and the outside-node-set share is the standardized share of the population within 200 km that lives in counties belonging to no sample area. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws). The principal-state row gives the same test with the 12 principal states as clusters. The difference row is the factor-adjusted minus the raw WRLURI coefficient from one regression stacking both networks, clustered by area.

OA.6.3 Where regulated metropolitan areas draw their dependence from

The metropolitan system can ask a question the state system cannot pose at all. When a regulated market is externally connected, is it connected to its neighbors inside the same state, or to markets elsewhere? Connectedness concentrated among same-state metropolitan areas would mean the state-scale results largely record tight internal clusters, and that aggregating a state's markets into one index measures composition rather than dependence. Connectedness running to markets in other states would mean the reverse.

Table OA.13, Part 1, splits each metropolitan area's external connectedness by the principal state of the source, same state or other states, and re-runs the regression on the log of each component. Two features of the decomposition govern how it must be read. Row normalization pins the total near 100, so the total varies little across areas (its within-year cross-sectional standard deviation is 4.6 points) while the two components vary by 14.6 and 14.8 among the areas that have both. Almost all of the cross-sectional variation is in the split. And that split depends in large part on how many sample metropolitan areas happen to share an area's state, a feature of the node set rather than of the housing market. Only principal-state effects absorb it, and the table reports those columns. Eighteen of the 100 areas have no same-state sample partner, so the component regressions use the other 82 and the total regression all 100.

In the raw network a one-standard-deviation increase in regulation is associated with 3.9 percent more connectedness to markets in other states and 12.4 percent less to markets within the same state (column 1). Regulated areas therefore take only slightly more of their forecast-error variance from elsewhere in total, but much more from other states and less from their own. Beyond the national cycle the other-state association is 4.1 percent across all areas and 5.0 percent among the 64 areas in states with at least three sample metropolitan areas, and the same-state one -8.2 and -10.6 (columns 2 and 4). Among the 64 areas the purge changes neither component by an amount distinguishable from zero. The split is by state, not by distance, and many nearby partners, such as New York and Bridgeport or Boston and Providence, count as other states.

The within-state comparison leans on a few areas. Twenty-one of the 100 areas have counties in more than one state, and they include several of the most regulated, among them Boston, Providence, Philadelphia, New York and Washington, D.C. Their nearest partners count as other states, while their same-state partners, upstate New York or Richmond and Norfolk, are distant, so the split can reflect where state lines fall. Part 2 of the table holds this fixed in two ways. With the share of an area's 2010 housing units outside its principal state added, the raw coefficients are -9.8 percent for the same-state component and 3.3 for the other-state one, and the factor-adjusted ones -5.4 and 3.1, with wild bootstrap p-values of 0.11 and 0.10. With the areas in more than one state dropped, the raw coefficients are -7.2 and 3.5 and the factor-adjusted ones -4.5 and 3.4, with p-values of 0.05 and 0.12. The split keeps its sign under both checks in both networks and stays precise in the raw network. Beyond the national cycle, between a third and a half of the same-state coefficient and up to a quarter of the other-state one rest on the areas that straddle state lines.

Table OA.13: External connectedness of metropolitan areas split by the source's state, and the areas that straddle a state line, raw and factor-adjusted networks, 1985–2025

Part 1. The split, with the controls of Table OA.12

	All areas		States with ≥ 3 metros	
	Raw	Factor-adjusted	Raw	Factor-adjusted
	(1)	(2)	(3)	(4)
WRLURI 2006 coefficient, dependent variable the log of the row's outcome				
External connectedness, total	0.45*	0.14	0.52***	0.35
	(0.19)	(0.31)	(0.14)	(0.29)
Component from same-state metros	−12.36***	−8.18*	−13.78***	−10.63***
	(3.77)	(3.66)	(3.93)	(3.63)
Component from other states' metros	3.92***	4.08*	4.50***	4.97**
	(1.06)	(1.73)	(1.02)	(1.78)
Controls of Table OA.12	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Principal-state fixed effects	✓	✓	✓	✓
Areas, total row / component rows	100 / 82	100 / 82	64 / 64	64 / 64
Sample	All areas	All areas	≥ 3 metros	≥ 3 metros

Note: Part 1: the 100 largest metropolitan areas, 41 annual networks 1985–2025; columns (1) and (3) raw, (2) and (4) factor-adjusted. Each area's external connectedness is split by the principal state of the source into the part from areas of the same principal state and the part from areas of other states; the split is by state, not by distance. Each cell is the WRLURI coefficient of its own regression of the log of the row's variable on the regressors, year effects and principal-state effects of Table OA.12, column 3, on all areas (columns 1–2) or on the areas of the principal states with at least three sample areas (columns 3–4). The 18 areas with no same-state sample partner have a same-state component of zero by design, so in columns (1) and (2) the component regressions use the other 82 areas and the total regressions all 100. Among the 82, the same-state and other-state components average 24.5 and 67.4 percentage points in the raw network and 32.6 and 54.8 in the factor-adjusted one. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

Table OA.13 (continued). Part 2, holding the areas that straddle a state line fixed

	Raw			Factor-adjusted		
	(1)	(2)	(3)	(4)	(5)	(6)
Coefficient on WRLURI 2006, with the log of each row's component as the dependent variable						
Panel A. Areas with a same-state sample partner						
Component from same-state metros	−12.36***	−9.77***	−7.18***	−8.18*	−5.35	−4.53**
	(3.77)	(2.78)	(1.99)	(3.66)	(2.59)	(1.83)
Component from other states' metros	3.92***	3.32***	3.45***	4.08*	3.05*	3.44
	(1.06)	(0.88)	(1.00)	(1.73)	(1.49)	(1.70)
Metropolitan areas	82	82	67	82	82	67
Panel B. States with at least three sample areas						
Component from same-state metros	−13.78***	−9.10**	−6.20***	−10.63***	−6.07	−3.98
	(3.93)	(3.12)	(2.05)	(3.63)	(2.86)	(1.96)
Component from other states' metros	4.50***	3.65***	3.53***	4.97**	3.37	3.33
	(1.02)	(0.99)	(1.09)	(1.78)	(1.66)	(1.88)
Metropolitan areas	64	64	52	64	64	52
Share of housing outside the principal state	–	✓	–	–	✓	–
Areas in more than one state dropped	–	–	✓	–	–	✓
Area controls	✓	✓	✓	✓	✓	✓
Year and principal-state fixed effects	✓	✓	✓	✓	✓	✓

Note: Part 2: columns (1) and (4) are the regressions of Part 1, each dependent variable the log of one component of the area's external connectedness. Columns (2) and (5) add the share of the area's 2010 housing units outside its principal state, standardized across the 100 areas; columns (3) and (6) drop the areas whose counties lie in more than one state. Every column has the area controls of Part 1 with year and principal-state effects. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.7 Migration and the Allocation of Connectedness

The tables of this section that extend Table 4, the specification sequence, the target-supply interactions and the parts and radius bounds of the metropolitan allocation, use its geography controls, in which log distance, contiguity (a shared Census division at metropolitan scale) and same Census region enter with their pairwise interactions. The full set of allocation specifications (Table OA.17), the gravity benchmarks and the fixed-pool comparison enter the three controls without their interactions.

OA.7.1 The specification sequence

Table OA.14 reports the sequence of designs behind Table 4, on the raw networks in Panel A and the factor-adjusted ones in Panel B: the pooled slope, the geography slope, geography with source-by-year effects, geography with node-year effects and node-year effects without geography, the last two being the designs of Table 4, which estimates them on all metropolitan pairs with a level of their own for each target-year's same-state sources (Table OA.15, below). On the raw networks, in which the national cycle is part of every cell, the state allocation association starts stronger and follows the same pattern as in the factor-adjusted networks. Across states the raw association begins at a pooled slope of 0.103 and a geography slope of 0.104, or 0.35 percentage points of conditional connectedness per standard deviation of exposure. Source-by-year and target-by-year effects without geography leave 0.068 (0.014), and with geography it falls to −0.001 and −0.014, as distance takes it over, with a coefficient that goes from 0.07 (0.15) to −0.20 (0.10). Among metropolitan pairs in different states the raw association is weaker than the factor-adjusted one, 0.056 pooled and 0.047 with geography. That is what one would expect if the common factor adds dependence to every pair whether or not its members are linked, so that the purge sharpens the migration association there. Source-year effects take the raw estimate to 0.018 and target-year effects to 0.009, no longer distinguishable from zero, while node-year effects without geography leave 0.046 (0.012). Across all metropolitan pairs the raw slope is 0.060 in column (4).

The allocation results move more with the own-lag treatment of Section 6.1 than the sending side does, plausibly because each step thins the cross-state network and leaves fewer links for migration to explain. In the raw networks the geography slope is 0.104 under the benchmark and 0.072 with the own lags free, and the unordered-pair-and-year slope 0.093 and 0.054, each in conditional-connectedness percentage points per point of migration exposure, and in the factor-adjusted networks they are smaller still. The designs with unordered-pair effects, which ask whether dependence has a direction that migration lines up with, have little to work with. More than 98 percent of the variance of exposure lies between links, unordered-pair-and-year effects retain 26 percent of it and the saturated effects, which add source-by-year and target-by-year effects to them, 14 percent. The designs therefore cannot tell whether migration makes dependence run more in one direction than the other, as when migration's association with metropolitan spillovers is absorbed by origin and destination traits (Malone, 2017). They follow the template of Landier et al. (2017), whose banking

regressor varies within pairs over time. Across states unordered-pair and year effects give 0.093 raw and 0.026 factor-adjusted, although the fall, −0.066 with a delete-one-node jackknife standard error of 0.050, is not itself distinguishable from zero. The saturated least-squares estimate is −0.010 raw and −0.003 factor-adjusted, its Poisson pseudo-maximum-likelihood counterpart −0.018 (0.009), and among metropolitan pairs in different states it is 0.008 raw and 0.011 factor-adjusted (Table OA.17).

Table OA.15 reports the metropolitan regressions of Table 4 on the factor-adjusted (Panel A) and the raw (Panel B) networks, with source-by-year and target-by-year effects in column (1) and the geography added on top in column (2). Both give the same-state sources of each target-year a level of their own. A single level for every target is a restrictive alternative, since targets have one to eleven same-state partners and split their within-state share among them, and its misfit loads on the exposure slope and on the distance slope through the node-year effects. With one level the factor-adjusted slope for pairs in different states is 0.094 (0.031) and the distance coefficient for pairs in different regions 0.25 (0.10), and interacting distance with the same-state indicator changes neither. With the target-specific level the slope is 0.084 (0.020) without geography and 0.067 (0.022) with it, beside the 0.060 of the out-of-state subsample, the distance coefficient is 0.01 (0.04) for pairs in different regions and −0.25 (0.07) within a division, and the slope among same-state pairs is −0.00 (0.03) without geography and −0.02 (0.03) with it, so there is no exposure gradient among them once each target's within-state share is its own. The all-pairs slopes without any same-state term, 0.060 raw and 0.171 factor-adjusted, are the within-state level loading on the exposure slope. In the raw network the target-specific slope for pairs in different states is 0.050 (0.010) without geography and 0.012 (0.007) with it.

Table OA.14: The specification sequence behind Table 4, raw and factor-adjusted networks, 2005–2025

Dependent variable	Conditional connectedness share (percentage points)				
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw networks					
States					
Conditional migration exposure	0.103***	0.104**	−0.001	−0.014	0.068***
	(0.027)	(0.044)	(0.021)	(0.020)	(0.014)
Log distance		0.069	−0.204**	−0.343***	
		(0.152)	(0.100)	(0.111)	
Metropolitan areas, out-of-state pairs					
Conditional migration exposure	0.056**	0.047*	0.018	0.009	0.046***
	(0.025)	(0.027)	(0.011)	(0.010)	(0.012)
Log distance		0.140***	−0.039	−0.065**	
		(0.037)	(0.025)	(0.030)	
Metropolitan areas, all pairs					
Conditional migration exposure	0.160***	0.071**	0.067***	0.060***	0.185***
	(0.036)	(0.027)	(0.023)	(0.022)	(0.033)
Log distance		0.145***	−0.042	−0.097**	
		(0.036)	(0.030)	(0.041)	
Panel B. Factor-adjusted networks					
States					
Conditional migration exposure	0.058***	0.039	0.006	0.000	0.059***
	(0.019)	(0.025)	(0.019)	(0.016)	(0.017)
Log distance		0.096	−0.089	−0.159*	
		(0.080)	(0.068)	(0.089)	
Metropolitan areas, out-of-state pairs					
Conditional migration exposure	0.077***	0.069***	0.070**	0.060**	0.077***
	(0.021)	(0.022)	(0.027)	(0.025)	(0.024)
Log distance		0.049*	−0.014	0.002	
		(0.027)	(0.038)	(0.041)	
Metropolitan areas, all pairs					
Conditional migration exposure	0.346***	0.148***	0.184***	0.171***	0.425***
	(0.066)	(0.046)	(0.057)	(0.056)	(0.072)
Log distance		0.068**	−0.020	−0.105	
		(0.028)	(0.058)	(0.088)	
Geography	–	✓	✓	✓	–
Source-by-year effects	–	–	✓	✓	✓
Target-by-year effects	–	–	–	✓	✓
Directed pair-years, states	47,376	47,376	47,376	47,376	47,376
metropolitan, out-of-state pairs	200,256	200,256	200,256	200,256	200,256
metropolitan, all pairs	207,900	207,900	207,900	207,900	207,900

Note: The 21 annual benchmark networks 2005–2025, raw in Panel A and factor-adjusted in Panel B. The dependent variable, the regressor, the samples and the geography controls are those of Table 4, whose two designs are columns (4) and (5) here, estimated there on all metropolitan pairs with the same-state terms; the row under each exposure coefficient is the log-distance coefficient where geography enters, the distance slope for pairs that are neither contiguous nor in the same region. Raw conditional shares average 0.88 percentage points across pairs of areas in other states and factor-adjusted ones 0.71. In each annual network a Poisson pseudo-maximum-likelihood regression of the pooled state-to-state flows on the three geography variables with origin and destination effects has a McFadden pseudo-R^2 of 0.93 to 0.94, against 0.62 to 0.67 for the effects alone, and geography accounts for 59 percent of the variance of state conditional migration exposure within source-by-year and target-by-year effects. Shared-node standard errors in parentheses, which allow two directed pairs to be dependent whenever they share a node; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against Student's t with one degree of freedom fewer than the number of nodes.

Table OA.15: The allocation regression on all metropolitan pairs with a same-state level for each target-year, factor-adjusted and raw networks, 2005–2025

Dependent variable	Conditional connectedness share (pp)	
	(1)	(2)
Panel A. Factor-adjusted network		
Exposure, pairs in different states	0.084^{***}	0.067^{***}
	(0.020)	(0.022)
Exposure, same-state pairs	-0.003	-0.024
	(0.026)	(0.032)
Log distance		0.009
		(0.036)
Log distance × contiguity		-0.252^{***}
		(0.072)
Log distance × same region		-0.052
		(0.049)
Log distance, same-state pairs		-0.625^{*}
		(0.329)
Panel B. Raw network		
Exposure, pairs in different states	0.050^{***}	0.012^{*}
	(0.010)	(0.007)
Exposure, same-state pairs	-0.002	-0.013
	(0.011)	(0.015)
Log distance		-0.061^{**}
		(0.027)
Log distance × contiguity		-0.139^{**}
		(0.063)
Log distance × same region		-0.119^{***}
		(0.044)
Log distance, same-state pairs		-0.267
		(0.164)
Other geography controls	–	✓
Source-by-year effects	✓	✓
Target-by-year effects	✓	✓
Target-by-year × same-state effects	✓	✓
Directed pair-years	207,522	207,522

Note: The 100 metropolitan areas in the 21 annual networks 2005–2025, all 9,900 directed pairs, of which the 7,644 pair-years of areas with the same principal state carry the same-state terms. Both columns give the same-state pairs a target-by-year effect of their own, which fits the 378 pair-years of the 18 areas with a single same-state partner exactly; column (1) has source-by-year and target-by-year effects, and column (2) adds the geography of Table 4 on top. In Panel A the two columns are columns (3) and (4) of that table. The rows report the slopes for pairs in different states and for same-state pairs. Shared-node standard errors in parentheses, which allow two directed pairs to be dependent whenever they share a node; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against Student's t with one degree of freedom fewer than the number of nodes.

OA.7.2 Same-state pairs and the target's own constraints

The metropolitan exposure coefficient is insensitive to how exposure is built. Re-estimated under three pool rules and both normalizations, including the one that routes the 39 percent of migrating exemptions with one end outside the system through an absorbing node, the saturated column ranges from 0.018 to 0.027 in the raw network and from 0.025 to 0.042 in the factor-adjusted one, and the rolling and fixed pools correlate at 0.99 across endpoints. The sensitivity that matters is to pairs of metropolitan areas with the same principal state, which the state panel does not contain. They are 3.7 percent of dyads, carry 19 percent of the variance of exposure (mean 5.5 percentage points against −0.2 for other pairs) and 39 percent of the variance of conditional shares. Table OA.14 therefore reports the metropolitan estimates with and without them, and Table 4 keeps every pair with a level of its own for each target-year's same-state sources (Table OA.15). In the raw network the saturated coefficient of 0.019 across all pairs (Table OA.17, column 5) is a mixture. The same-state slope is 0.028 (0.010), and restricting the panel to out-of-state pairs leaves 0.008 (0.009), an estimate as small and as indistinguishable from zero as the state panel's −0.010 (0.018). Metropolitan areas in the same state share state-level shocks, institutions and lenders as well as migrants, and the allocation design cannot separate those channels for in-state dyads.

The same channel carries the heterogeneity by regulation. Table OA.16 interacts conditional migration exposure with the target's standardized regulation and steep land share under the geography and node-year design of Table 4, with that table's interacted geography controls. The direct weights of Section 4 imply no interaction in conditional shares, because the supply multiplier $\rho_{i,t}$ scales every weight of a target's row and cancels from conditional weights, so the table asks whether allocation departs from that prediction. Among metropolitan pairs in different states it does not. The regulation interaction is 0.007 (0.015) in the factor-adjusted network and 0.003 (0.007) in the raw one, the slope interaction 0.003 (0.013) and 0.000 (0.006), and on the apportioned node set of Table OA.10 the factor-adjusted interactions are −0.004 (0.010) and −0.003 (0.008). Across states they are −0.001 (0.008) and 0.019 (0.012) in the factor-adjusted network and 0.015 (0.009) and 0.007 (0.011) in the raw one.

Across all metropolitan pairs the regulation interaction is negative, −0.063 (0.021) in the factor-adjusted network and −0.038 (0.009) in the raw one, and −0.057 (0.035) on the apportioned node set. Read on its own, that says regulated markets follow their migration linkages less closely. The destination split of Section OA.6.3 offers a reading of it. Same-state pairs are the pairs with the highest exposure, and regulated metropolitan areas place less of their dependence on exactly those partners and more on markets in other states. Over all pairs, therefore, the high-exposure same-state partners of a regulated target receive less connectedness than exposure alone would predict, and the exposure gradient flattens with regulation. The pattern is consistent with in-state clusters rather than with migration-linked sources in general, a conjecture rather than an estimate, and it vanishes once same-state pairs are excluded.

Table OA.16: Allocation and the target's supply constraint, column (4) of Table 4, 2005–2025

		Benchmark, 100 largest		Apportioned, 100	
	States	All pairs	Out of state	All pairs	Out of state
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw networks					
Exposure, Table 4 column (4)	−0.014	0.060***	0.009	0.067***	0.018***
	(0.020)	(0.022)	(0.010)	(0.018)	(0.007)
Exposure	−0.016	0.063***	0.008	0.067***	0.017***
	(0.019)	(0.020)	(0.010)	(0.019)	(0.006)
Exposure × target WRLURI	0.015	−0.038***	0.003	−0.023*	0.003
	(0.009)	(0.009)	(0.007)	(0.013)	(0.004)
Exposure × target steep land share	0.007	0.004	0.000	−0.002	0.005
	(0.011)	(0.009)	(0.006)	(0.012)	(0.003)
Pair-years	47,376	207,900	200,256	205,821	202,125
Panel B. Factor-adjusted networks					
Exposure, Table 4 column (4)	0.000	0.171***	0.060**	0.204***	0.053***
	(0.016)	(0.056)	(0.025)	(0.047)	(0.013)
Exposure	−0.002	0.176***	0.059**	0.202***	0.054***
	(0.011)	(0.052)	(0.025)	(0.045)	(0.013)
Exposure × target WRLURI	−0.001	−0.063***	0.007	−0.057*	−0.004
	(0.008)	(0.021)	(0.015)	(0.035)	(0.010)
Exposure × target steep land share	0.019	0.002	0.003	−0.015	−0.003
	(0.012)	(0.019)	(0.013)	(0.030)	(0.008)
Pair-years	47,376	207,900	200,256	205,821	202,125

Note: Dependent variable: the conditional connectedness share, 21 annual networks 2005–2025, each node set's own networks and migration exposure. Every column has source-by-year and target-by-year effects and the interacted geography controls of Table 4. The first row of each panel is the exposure coefficient under the geography and node-year design without the same-state terms, as in Table OA.14, column (4); the next three rows are one regression that adds exposure times the target's WRLURI 2006 and steep land share, each standardized over the node set's targets, whose main effects the target-by-year effects absorb. The apportioned node set is that of Table OA.10; Cheyenne, Wyoming, has no WRLURI respondent, so columns (4) and (5) drop it as a target and keep it as a source. Shared-node standard errors in parentheses, which allow two directed pairs to be dependent whenever they share a node; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against Student's t with one degree of freedom fewer than the number of nodes.

Table OA.17 reports the full set of allocation specifications at both scales in the raw and the factor-adjusted networks, with the three geography controls (log distance, contiguity or a shared Census division, and same Census region) entered without interactions, including the target-supply interactions under unordered-pair-and-year effects. Under those effects the target's own regulation and steep land share vary within a pair because the two directions of a pair have different targets, so the column includes them as main effects. No interaction is then distinguishable from zero at either scale in the raw network. The regulation interaction is −0.022 (0.019) across states and −0.008 (0.006) and −0.006 (0.009) across all and out-of-state metropolitan pairs, and the slope interaction 0.007 (0.006), 0.007 (0.005) and 0.002 (0.006).

Table OA.17: Migration exposure, target supply and the allocation of connectedness across sources, raw and factor-adjusted networks, 2005–2025

Dependent variable	Conditional connectedness share (percentage points)				
	(1)	(2)	(3)	(4)	(5)
Raw networks					
States (48 targets, 47 sources)					
Migration exposure	0.104**	0.093**	0.079**	−0.008	−0.010
	(0.042)	(0.044)	(0.039)	(0.020)	(0.018)
	[0.054]	[0.053]	[0.051]	[0.027]	[0.023]
× target WRLURI 2006			−0.022	0.020**	
			(0.019)	(0.008)	
			[0.026]	[0.011]	
× target steep land share			0.007	0.003	
			(0.006)	(0.009)	
			[0.013]	[0.015]	
Metropolitan areas, all pairs (100 targets, 99 sources)					
Migration exposure	0.107***	0.035**	0.027*	0.104***	0.019***
	(0.033)	(0.017)	(0.016)	(0.026)	(0.006)
	[0.043]	[0.023]	[0.021]	[0.035]	[0.008]
× target WRLURI 2006			−0.008	−0.036***	
			(0.006)	(0.009)	
			[0.008]	[0.015]	
× target steep land share			0.007	−0.005	
			(0.005)	(0.010)	
			[0.007]	[0.014]	
Metropolitan areas, out-of-state pairs					
Migration exposure	0.052**	0.041	0.026	0.016	0.008
	(0.026)	(0.035)	(0.031)	(0.011)	(0.009)
	[0.035]	[0.049]	[0.046]	[0.015]	[0.014]
× target WRLURI 2006			−0.006	0.003	
			(0.009)	(0.007)	
			[0.014]	[0.010]	
× target steep land share			0.002	−0.004	
			(0.006)	(0.006)	
			[0.012]	[0.009]	
Geography	✓	–	–	✓	–
Year fixed effects	✓	✓	✓	–	–
Unordered-pair fixed effects	–	✓	✓	–	✓
Source-by-year and target-by-year fixed effects	–	–	–	✓	✓
Target main effects	–	–	✓	–	–
Directed pair-years, states	47,376	47,376	47,376	47,376	47,376
metropolitan, all pairs	207,900	207,900	207,900	207,900	207,900
metropolitan, out-of-state pairs	200,256	200,256	200,256	200,256	200,256
Within R^2					
States	0.0348	0.0092	0.0233	0.0479	0.0001
Metropolitan, all pairs	0.2048	0.0046	0.0402	0.2803	0.0014
Metropolitan, out-of-state pairs	0.0327	0.0048	0.0555	0.0452	0.0002

Note: 21 annual benchmark networks 2005–2025; this part raw, the continuation factor-adjusted. The dependent variable, the regressor, the samples and the shared-node standard errors and stars are those of Table 4, with the three geography controls entered without interactions; brackets contain delete-one-node jackknife standard errors. Column (1) has geography and year effects, column (2) unordered-pair-and-year effects and column (5) source-by-year, target-by-year and unordered-pair effects, the saturated design. Columns (3) and (4) add the interactions of exposure with the target's standardized WRLURI 2006 and steep land share, under unordered-pair-and-year effects in column (3), where the two target main effects enter as well, and under node-year effects with geography in column (4), where they are absorbed.

Table OA.17 (continued). Factor-adjusted networks

Dependent variable	Conditional connectedness share (percentage points)				
	(1)	(2)	(3)	(4)	(5)
Factor-adjusted networks					
States (48 targets, 47 sources)					
Migration exposure	0.044*	0.026	0.015	0.009	−0.003
	(0.024)	(0.023)	(0.026)	(0.012)	(0.017)
	[0.030]	[0.029]	[0.036]	[0.020]	[0.023]
× target WRLURI 2006			−0.020	0.005	
			(0.016)	(0.007)	
			[0.024]	[0.012]	
× target steep land share			0.007	0.016	
			(0.011)	(0.011)	
			[0.019]	[0.019]	
Metropolitan areas, all pairs (100 targets, 99 sources)					
Migration exposure	0.232***	0.035**	0.028**	0.277***	0.025*
	(0.059)	(0.013)	(0.013)	(0.064)	(0.014)
	[0.078]	[0.017]	[0.017]	[0.088]	[0.018]
× target WRLURI 2006			0.014*	−0.057**	
			(0.008)	(0.023)	
			[0.013]	[0.042]	
× target steep land share			−0.006	−0.021	
			(0.008)	(0.021)	
			[0.011]	[0.034]	
Metropolitan areas, out-of-state pairs					
Migration exposure	0.074***	0.035*	0.026	0.069**	0.011
	(0.021)	(0.019)	(0.017)	(0.026)	(0.014)
	[0.030]	[0.025]	[0.025]	[0.042]	[0.021]
× target WRLURI 2006			0.011	0.008	
			(0.011)	(0.016)	
			[0.018]	[0.025]	
× target steep land share			−0.002	−0.002	
			(0.007)	(0.012)	
			[0.011]	[0.021]	
Geography	✓	–	–	✓	–
Year fixed effects	✓	✓	✓	–	–
Unordered-pair fixed effects	–	✓	✓	–	✓
Source-by-year and target-by-year fixed effects	–	–	–	✓	✓
Target main effects	–	–	✓	–	–
Directed pair-years, states	47,376	47,376	47,376	47,376	47,376
metropolitan, all pairs	207,900	207,900	207,900	207,900	207,900
metropolitan, out-of-state pairs	200,256	200,256	200,256	200,256	200,256
Within R^2					
States	0.0101	0.0005	0.0071	0.0127	0.0000
Metropolitan, all pairs	0.3012	0.0034	0.0112	0.3427	0.0013
Metropolitan, out-of-state pairs	0.0309	0.0030	0.0145	0.0259	0.0002

Note: Factor-adjusted benchmark networks; construction, samples and inference as in the first part of the table. Conditional shares average 0.71 percentage points across out-of-state pairs.

OA.7.3 Gravity benchmarks

Table OA.18 compares gravity and unordered-pair specifications at state scale, including models without migration exposure, in the raw network (Panel A) and the factor-adjusted one (Panel B). The fit increment measures how much adding exposure improves fit with the same controls. The rows with unordered-pair and source-by-year effects and with unordered-pair and target-by-year effects isolate the contribution of each set of node-year effects, and the main allocation table includes both sets together. Migration's association with the allocation is less than half as large in the factor-adjusted network, 0.044 (0.024) with geography against 0.104 (0.042), and 0.026 (0.023) with unordered-pair-and-year effects against 0.093 (0.044). The incremental fit of exposure over geography falls from 0.020 to 0.003. With the interacted geography controls the two geography slopes are 0.104 (0.044) and 0.039 (0.025).

Table OA.18: Geography benchmarks and the incremental association of migration exposure, raw and factor-adjusted networks, 2005–2025

Design	Estimator	Exposure	Shared-node SE	Node JK SE	Fit	Incremental fit
Panel A. Raw network						
Geography and year, no exposure	OLS	–	–	–	0.0147	–
Geography and year	OLS	0.104	0.042	0.054	0.0348	0.0201
Unordered-pair and year, no exposure	OLS	–	–	–	0.0000	–
Unordered-pair and year	OLS	0.093	0.044	0.053	0.0092	0.0092
Unordered-pair and source-by-year	OLS	−0.010	0.018	0.023	0.0001	0.0001
Unordered-pair and target-by-year	OLS	−0.010	0.018	0.023	0.0001	0.0001
Geography and year, no exposure	PPML	–	–	–	0.0062	–
Geography and year	PPML	0.076	0.026	0.034	0.0131	0.0069
Unordered-pair and year, no exposure	PPML	–	–	–	0.0991	–
Unordered-pair and year	PPML	0.073	0.031	0.037	0.1015	0.0024
Unordered-pair and source-by-year	PPML	−0.018	0.009	0.013	0.2659	0.0001
Unordered-pair and target-by-year	PPML	−0.018	0.010	0.013	0.1463	0.0001
Panel B. Factor-adjusted network						
Geography and year, no exposure	OLS	–	–	–	0.0074	–
Geography and year	OLS	0.044	0.024	0.030	0.0101	0.0027
Unordered-pair and year, no exposure	OLS	–	–	–	0.0000	–
Unordered-pair and year	OLS	0.026	0.023	0.029	0.0005	0.0005
Unordered-pair and source-by-year	OLS	−0.003	0.017	0.023	0.0000	0.0000
Unordered-pair and target-by-year	OLS	−0.002	0.017	0.024	0.0000	0.0000
Geography and year, no exposure	PPML	–	–	–	0.0039	–
Geography and year	PPML	0.036	0.018	0.022	0.0052	0.0013
Unordered-pair and year, no exposure	PPML	–	–	–	0.0939	–
Unordered-pair and year	PPML	0.023	0.021	0.027	0.0941	0.0002
Unordered-pair and source-by-year	PPML	−0.004	0.013	0.019	0.1551	0.0000
Unordered-pair and target-by-year	PPML	−0.003	0.014	0.019	0.1093	0.0000

Note: Geography and unordered-pair specifications of the state allocation regression, with and without conditional migration exposure; Panel A raw, Panel B factor-adjusted, with exposure, geography (log distance, contiguity and same Census region, without interactions), sample (47,376 directed pair-years) and fixed effects identical. Rows marked no exposure omit the regressor. PPML entries are average partial associations, the index coefficient times 100/47. Fit is the within R^2 for OLS and the pseudo-R^2 for PPML, and incremental fit compares models with the same fixed effects and geography; fit differences are descriptive. The standard-error columns report shared-node and delete-one-node jackknife standard errors, holding each estimated network fixed.

OA.7.4 Fixed and rolling migration pools

Table OA.19 fixes migration exposure at its 1990–1994 value for every endpoint and compares it with the rolling five-year pools, in the raw network and with the three geography controls entered without interactions. Connectedness outcomes, covariates and the 48-state universe are unchanged. This checks whether the allocation results depend on updating migration weights; it does not use a shorter averaging period.

Table OA.19: Conditional allocation: rolling versus fixed migration pools, 2005–2025

Model	Estimator	Rolling	Shared-node SE	Fixed 1990–94	Shared-node SE
Geography and year	OLS	0.104	0.042	0.105	0.039
Unordered-pair and year	OLS	0.093	0.044	0.096	0.041
Source-year and target-year	OLS	0.068	0.014	0.068	0.013
Source-year and target-year	PPML	0.053	0.010	0.052	0.010
Node-year and geography	OLS	−0.004	0.019	−0.005	0.018
Node-year and geography	PPML	−0.012	0.011	−0.014	0.011
Node-year and pair	OLS	−0.010	0.018	−0.010	0.018
Node-year and pair	PPML	−0.018	0.009	−0.019	0.009

Note: Conditional connectedness and migration exposure in percentage points across a target's 47 external sources, exposure centered at 100/47; 47,376 directed pair-years. The rolling columns use IRS pools from $t-15$ through $t-11$ and the fixed columns the 1990–94 pool at every endpoint. Geography is log distance, contiguity and same Census region, and pair effects are unordered. Shared-node standard errors beside each estimate.

OA.7.5 The unconditional cell

The direct weights of Section 4 imply an interaction between exposure and the target's supply constraint, but on the unconditional cell rather than the conditional share. Writing $W_{ij,t} = \rho_{i,t}\psi_{ij,t}$ (equation 9), the conditional share divides by the row total and $\rho_{i,t}$ cancels exactly, whereas the unconditional cell keeps it. Regressing $C_{ij,t}$ on diagonal-inclusive exposure, its interaction with target regulation and its interaction with target openness, under source-by-year and target-by-year effects, returns an interaction of 0.325 with a bootstrap standard error of 0.078, from 999 draws that resample states together with every directed pair in which a drawn state appears. That estimate is an accounting identity rather than evidence. Since $C_{ij} = F_i\bar{C}_{ij}/100$, the cross-derivative decomposes into a scaling term, the receiving-side gradient times the allocation slope, and a term in the conditional interaction. On the same 21 endpoints the two terms are 0.025 and 0.299 and sum to 0.324 against the 0.325 estimated directly; the gap, 0.001 with a standard error of 0.008 from the same draws, is approximation error. The log form carries more content. Poisson pseudo-maximum-likelihood estimation of $C_{ij,t}$ on $\log \psi_{ij,t}$ with the same effects gives an elasticity of 0.134 (0.012) against a direct-weight value of exactly one, which no bootstrap draw reaches. Indirect paths through intermediate markets (the Neumann series of Appendix OA.9.6) and the innovation covariance smooth the mapping from W_{ij} to C_{ij}, so an elasticity below one is expected, and the ordering restriction—that unconditional dependence rises with exposure—holds; the paper claims no more than the ordering. The components are products and ratios of regression coefficients, not

single coefficients, so this accounting keeps its paired node-cluster bootstrap rather than the wild bootstrap-t of the regression tables, and the calculations are on the raw network.

OA.7.6 Direct and common-origin parts of migration exposure, and pairs of areas close to each other

Equation (8) splits exposure into residents of the target who move to the source, residents of the target who came from the source weighted by the share of the source's residents who stay, and the origins the two share. Table OA.20, Part 1, reports the three parts as shares of off-diagonal exposure, averaged over the 21 endpoints and with their range. In the state matrices, which keep stayers on the diagonal, the two direct parts account for 48.2 and 50.5 percent on average and the common-origin part for 1.3 percent, between 1.2 and 1.4 at every endpoint, and conditional exposure built from the direct parts alone correlates at 0.99998 with the full measure. The metropolitan matrices hold movers only, so an area's diagonal counts moves among its own counties and the common-origin part is 39.1 percent across all pairs and 42.8 percent across pairs in different states.

Part 2 of the table repeats the specification sequence of Table OA.14 with the direct part alone and with the direct and common-origin parts entered together, each built as conditional exposure is built, so that the two add up to it. Among metropolitan pairs in different states, beyond the national cycle, the direct part carries the association in column (4), 0.064 (0.029), and the common-origin part is 0.013 (0.050). Across states the two parts cannot be told apart. With both entered, the raw geography slope loads on the common-origin part, 15.1 (5.6), whose standard deviation is 0.025 percentage points, but that coefficient is −9.2 in column (4) and 4.4 in column (5), and in the factor-adjusted network neither part is distinguishable from zero once geography enters.

Schubert (2024) excludes links within 50, 100 and 150 miles of each commuting zone. Part 3 of the table applies the same exclusion to the out-of-state metropolitan allocation of Table 4, dropping both directions of every pair whose internal points lie within the radius and keeping exposure as built. At 50 miles, which drops six directed pairs, the factor-adjusted column (4) estimate is 0.063 (0.026). At 100 miles, which drops 74, it is 0.047 (0.025), with an interval of $[-0.002, 0.095]$, and at 150 miles, which drops 188, 0.045 (0.028). The raw estimate falls from 0.009 to 0.002 at 100 miles. The factor-adjusted estimates of columns (1) to (3) keep intervals that exclude zero at every radius. Most of the association that survives node-year effects therefore remains without the nearby out-of-state pairs, but it is no longer distinguishable from zero.

Table OA.20: Direct and common-origin parts of migration exposure, the allocation ladder with each part, and the metropolitan allocation without nearby pairs, 2005–2025

Part 1. Shares of off-diagonal exposure, in percent

Endpoint	States			Metropolitan areas		
	Target's residents (D1)	Source's residents (D2)	Common origins (C)	Target's residents (D1)	Source's residents (D2)	Common origins (C)
Mean across endpoints	48.22	50.48	1.29	25.25	35.65	39.10
Range across the 21 endpoints	[47.01, 49.09]	[49.72, 51.58]	[1.19, 1.42]	[23.99, 25.97]	[35.16, 36.28]	[38.14, 40.76]
out-of-state pairs				25.17	32.05	42.77
Mean external openness (percent)		5.69			70.51	
Correlation of $x_{ij,t}$ with its direct part		0.99998			0.96926	

Note: Part 1: shares, in percent, of total off-diagonal migration exposure in each annual network, with $\Psi_t = \Phi_t M_t$ from IRS flows pooled over $t-15$ through $t-11$ (equation 7). Exposure splits exactly into three parts (equation 8): D1, the target's own residents who move to the source; D2, the target's residents who came from the source weighted by the source's diagonal share; and C, the common-origin part. The state flow matrix keeps the IRS same-state non-migrants on the diagonal; the metropolitan matrix, IRS county-to-county flows aggregated to CBSA pairs and normalized within the system, holds movers only, so an area's diagonal is the moves between its own counties. The out-of-state row restricts the metropolitan sums to pairs of areas with different principal states. The last row is the pooled correlation between conditional migration exposure and the same construction applied to the direct part alone, and the range row is the smallest and largest share across the 21 endpoints.

Table OA.20 (continued). Part 2, the allocation ladder with each part, raw network

Dependent variable	Conditional connectedness share (percentage points)				
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw network					
States					
Direct part alone	0.103***	0.104**	−0.001	−0.014	0.068***
	(0.027)	(0.044)	(0.022)	(0.020)	(0.014)
	[0.346]	[0.349]	[−0.003]	[−0.046]	[0.228]
Direct part, both entered	0.045**	−0.001	0.007	−0.001	0.059***
	(0.019)	(0.033)	(0.024)	(0.022)	(0.016)
	[0.150]	[−0.003]	[0.022]	[−0.003]	[0.198]
Common-origin part, both entered	12.02**	15.09***	−4.55	−9.16**	4.38
	(5.46)	(5.59)	(4.23)	(3.83)	(5.64)
	[0.302]	[0.379]	[−0.114]	[−0.230]	[0.110]
Metropolitan areas, out-of-state pairs					
Direct part alone	0.049*	0.035	0.017	0.010	0.046***
	(0.025)	(0.026)	(0.011)	(0.010)	(0.013)
	[0.072]	[0.052]	[0.025]	[0.014]	[0.067]
Direct part, both entered	0.007	−0.005	0.016	0.010	0.036***
	(0.019)	(0.019)	(0.011)	(0.011)	(0.012)
	[0.010]	[−0.008]	[0.023]	[0.014]	[0.052]
Common-origin part, both entered	0.263***	0.261***	0.038	−0.005	0.123**
	(0.061)	(0.065)	(0.039)	(0.040)	(0.051)
	[0.130]	[0.129]	[0.019]	[−0.003]	[0.061]
Metropolitan areas, all pairs					
Direct part alone	0.170***	0.065**	0.067**	0.060**	0.192***
	(0.042)	(0.029)	(0.025)	(0.025)	(0.038)
	[0.361]	[0.138]	[0.141]	[0.128]	[0.408]
Direct part, both entered	0.127***	0.041	0.063**	0.058**	0.159***
	(0.039)	(0.028)	(0.025)	(0.025)	(0.036)
	[0.270]	[0.086]	[0.133]	[0.123]	[0.337]
Common-origin part, both entered	0.314***	0.220***	0.102***	0.075**	0.363***
	(0.052)	(0.041)	(0.036)	(0.035)	(0.067)
	[0.212]	[0.149]	[0.069]	[0.051]	[0.245]

Note: Part 2: the 21 annual benchmark networks 2005–2025, raw here and factor-adjusted in the continuation. The dependent variable, samples and geography controls are those of Table 4, whose exposure regressor the two parts replace, and the columns follow the specification sequence of Table OA.14. The direct part and the common-origin part of conditional migration exposure are built as the regressor is, so the two add up to it exactly. Each block reports the direct part alone and then the two parts together; geography coefficients are not shown. Brackets contain standardized coefficients, the coefficient times the standard deviation of the part in the estimation sample (3.345 and 0.025 across states, 1.457 and 0.494 across out-of-state metropolitan pairs, 2.121 and 0.676 across all metropolitan pairs); the state common-origin coefficients are printed to two decimals because that part varies little. Shared-node standard errors in parentheses, which allow two directed pairs to be dependent whenever they share a node; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against Student's t with one degree of freedom fewer than the number of nodes.

Table OA.20 (continued). Part 2, factor-adjusted networks

Dependent variable	Conditional connectedness share (percentage points)				
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography
	(1)	(2)	(3)	(4)	(5)
Panel B. Factor-adjusted network					
States					
Direct part alone	0.059^{***}	0.039	0.006	0.000	0.059^{***}
	(0.019)	(0.025)	(0.019)	(0.016)	(0.017)
	[0.196]	[0.131]	[0.022]	[0.000]	[0.197]
Direct part, both entered	0.048^{***}	0.006	0.008	0.004	0.051^{***}
	(0.016)	(0.023)	(0.022)	(0.020)	(0.015)
	[0.159]	[0.020]	[0.026]	[0.014]	[0.171]
Common-origin part, both entered	2.27	4.73	−0.78	−2.97	3.83
	(3.41)	(3.88)	(4.00)	(3.95)	(3.49)
	[0.057]	[0.119]	[−0.020]	[−0.075]	[0.096]
Metropolitan areas, out-of-state pairs					
Direct part alone	0.086^{***}	0.074^{***}	0.074^{**}	0.065^{**}	0.086^{***}
	(0.026)	(0.026)	(0.030)	(0.028)	(0.028)
	[0.126]	[0.108]	[0.107]	[0.094]	[0.125]
Direct part, both entered	0.068^{**}	0.055^{*}	0.070^{**}	0.064^{**}	0.082^{***}
	(0.028)	(0.028)	(0.030)	(0.029)	(0.030)
	[0.099]	[0.080]	[0.102]	[0.093]	[0.120]
Common-origin part, both entered	0.115^{***}	0.127^{***}	0.073	0.013	0.040
	(0.042)	(0.043)	(0.057)	(0.050)	(0.056)
	[0.057]	[0.063]	[0.036]	[0.007]	[0.020]
Metropolitan areas, all pairs					
Direct part alone	0.394^{***}	0.166^{***}	0.190^{***}	0.178^{***}	0.451^{***}
	(0.082)	(0.057)	(0.063)	(0.062)	(0.084)
	[0.836]	[0.351]	[0.403]	[0.377]	[0.957]
Direct part, both entered	0.349^{***}	0.150^{**}	0.184^{***}	0.174^{***}	0.390^{***}
	(0.082)	(0.059)	(0.064)	(0.062)	(0.081)
	[0.741]	[0.318]	[0.389]	[0.369]	[0.827]
Common-origin part, both entered	0.328^{***}	0.140^{**}	0.184^{**}	0.138^{*}	0.661^{***}
	(0.098)	(0.055)	(0.074)	(0.072)	(0.139)
	[0.222]	[0.095]	[0.124]	[0.093]	[0.447]

Note: Factor-adjusted benchmark networks; samples, columns, regressors, standardization and inference as in Part 2 above.

Table OA.20 (continued). Part 3, out-of-state metropolitan pairs without nearby pairs of areas

Dependent variable	Conditional connectedness share (percentage points)					Directed
	Bivariate	Geography slope	Geography and source-by-year effects	Geography and node-year effects	Node-year effects without geography	pairs dropped
Exclusion radius	(1)	(2)	(3)	(4)	(5)	
Panel A. Out-of-state pairs, raw network						
No exclusion	0.056**	0.047*	0.018	0.009	0.046***	0
	(0.025)	(0.027)	(0.011)	(0.010)	(0.012)	
Within 50 miles	0.058**	0.049*	0.019*	0.009	0.047***	6
	(0.025)	(0.026)	(0.011)	(0.010)	(0.012)	
Within 100 miles	0.065**	0.058*	0.016	0.002	0.043***	74
	(0.030)	(0.030)	(0.014)	(0.014)	(0.016)	
Within 150 miles	0.070**	0.064**	0.016	−0.001	0.037**	188
	(0.031)	(0.031)	(0.017)	(0.018)	(0.018)	
Panel B. Out-of-state pairs, factor-adjusted network						
No exclusion	0.077***	0.069***	0.070**	0.060**	0.077***	0
	(0.021)	(0.022)	(0.027)	(0.025)	(0.024)	
Within 50 miles	0.079***	0.071***	0.074***	0.063**	0.080***	6
	(0.021)	(0.022)	(0.027)	(0.026)	(0.024)	
Within 100 miles	0.066***	0.065***	0.065**	0.047*	0.055**	74
	(0.019)	(0.020)	(0.029)	(0.025)	(0.024)	
Within 150 miles	0.065***	0.066***	0.066**	0.045	0.047*	188
	(0.020)	(0.020)	(0.030)	(0.028)	(0.027)	

Note: Part 3: each cell is the coefficient on conditional migration exposure in the specification sequence of Table OA.14 among out-of-state pairs, with exposure kept as built, after dropping both directions of every pair whose Census Gazetteer internal points lie within 50, 100 or 150 miles of each other; the last column counts the directed pairs dropped of the 9,536 out-of-state directed pairs in each annual network. In the factor-adjusted network at 100 miles, column (4) is 0.047 with a 95-percent interval from −0.002 to 0.095. Shared-node standard errors in parentheses, which allow two directed pairs to be dependent whenever they share a node; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against Student's t with one degree of freedom fewer than the number of nodes.

OA.8 Measurement and Influence

OA.8.1 Alternative aggregations of the constraint measures

Table OA.21 varies the aggregation of the regulation index on the same 48-state sample and the same model, in the raw network (columns 1–3) and the factor-adjusted one (columns 4–6). Survey weights and jurisdiction land area change the relative weights of the responding jurisdictions, and a series in the aggregation inputs labeled as weighted by metropolitan population is the equal-jurisdiction mean (Section OA.1.3) and is not listed. In the factor-adjusted network the TO association is 7.8 (2.2) with survey weights and 7.9 (2.4) with land-area weights, against 7.5 with equal jurisdictions, and it remains positive under every weighting rule. Panel B gives the range of the WRLURI coefficient across the seven aggregations of the steep land share described in Section OA.1.3, 7.45 to 7.52 in the factor-adjusted network, under none of which the land measure is distinguishable from zero at the 5-percent level.

Table OA.21: Regulation under alternative aggregations of the index, raw and factor-adjusted networks, 1985–2025

	Raw network			Factor-adjusted network		
	log TO (1)	log FROM (2)	log NET (3)	log TO (4)	log FROM (5)	log NET (6)
Panel A. Coefficient on WRLURI 2006 under each aggregation of the regulation index						
Equal jurisdictions (Table 2)	10.54**	2.76***	7.78**	7.50***	1.67	5.83**
	(3.62)	(0.60)	(3.40)	(2.32)	(1.05)	(2.08)
Survey weights	10.90**	2.72***	8.18**	7.78***	1.64	6.14**
	(3.44)	(0.59)	(3.22)	(2.18)	(1.05)	(2.06)
Jurisdiction land-area weights	9.74**	2.49***	7.25*	7.89***	1.15	6.74**
	(3.90)	(0.64)	(3.68)	(2.42)	(1.17)	(2.21)
Panel B. Range of the WRLURI 2006 coefficient across the seven aggregations of the steep land share						
Range	10.52–10.63	2.75–2.78	7.77–7.84	7.45–7.52	1.66–1.72	5.74–5.84

Note: 48 states, 41 annual networks 1985–2025; raw network in columns (1)–(3), factor-adjusted in (4)–(6). Each entry of Panel A comes from the regression of Table 2 with the regulation index replaced by the named aggregation, standardized across the 48 states; a further series in the aggregation inputs, labeled as weighted by metropolitan population, equals the equal-jurisdiction mean and is not reported. Panel B gives the range of the WRLURI 2006 coefficient over the seven regressions in which the steep land share is replaced in turn by its four weighting rules, the county slope comparator and the two boundary conventions of Section OA.1.3, under none of which the land coefficient is distinguishable from zero at the 5-percent level. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

OA.8.2 What the regulation index measures

Table OA.22, Part 1, decomposes the index into its survey-content and state-rating blocks across the 48 states, for log TO, log FROM and log NET in the raw network (columns 1–3) and the factor-adjusted one (columns 4–6). Two of the index's eleven components measure how involved a state's legislature and courts are in land use, and they vary only across states. The court index is

a rating of Foster and Summers (2005), and the political-involvement index combines that rating with the state mean of one survey answer (Gyourko et al., 2008); together they carry 16.5 percent of the index's loading mass. The index's own component weights are recovered by regressing it on its standardized components, which returns an R^2 of one and therefore the exact weighting, and each block enters with those weights, so that the blocks of a column add up to the published index before each is standardized. Panel A enters survey content and the ratings together, Panel B splits the survey content, and Panel C enters each block on its own. Across the 48 states the survey content and the local instruments correlate 0.98, and survey content and the ratings 0.67, which is why every block but supply restrictions is positively associated with external connectedness when it enters alone. For raw log FROM the survey-minus-ratings contrast is 3.20 (1.86), with a 95-percent interval of $[-0.86, 7.79]$ (column 2). Beyond the national cycle the survey content carries the receiving side, 3.25 (0.86) against -1.79 (1.43) for the ratings, a contrast of 5.04 (2.02) (column 5). The factor-adjusted sending side loads the other way. The ratings carry 7.63 (3.51) of log TO and 9.41 (3.32) of log NET, survey content 1.17 (2.56) and -2.08 (2.71), and the contrasts are -6.46 (5.55) and -11.49 (5.61), the second distinguishable from zero at the 10-percent level. The point estimates place the sending side in state institutions and the receiving side in local rules; outside the factor-adjusted state receiving side the blocks cannot be ranked with confidence.

Part 2 of the table repeats the decomposition at the metropolitan scale, for the receiving side, in the raw and the factor-adjusted networks. The State Court Involvement Index is identical to the published Foster–Summers Judicial Profile to machine precision, and is uncorrelated with the survey's own questions about court involvement, at -0.003 and -0.031. In the raw network, column (1), with division effects, separates the nine components with survey content from the two external ratings. Survey content is associated with 0.75 percent more external connectedness per standard deviation (0.25), while the ratings carry 0.17 (0.35). Column (2) splits the survey content into the seven local land-use instruments—project approval, zoning approval, approval delay, open space, density restriction, exactions and local assembly—local political pressure and supply restrictions. Local political pressure carries 0.64 (0.24) and is the one block distinguishable from zero. Column (3) repeats the survey split within states, where the ratings are absorbed. There no single survey block is distinguishable from zero, although the three are jointly ($p = 0.007$), and entered alone, survey content, the local instruments and local political pressure are each positively associated with external connectedness. Collinearity is moderate, with a largest variance inflation factor of 2.8, and the blocks are jointly distinguishable from zero in every column, but the hypothesis that the block associations are equal is not rejected in any column, at $p = 0.20$, 0.18 and 0.11. In the factor-adjusted network the block point estimates hardly move, the standard errors roughly double, and no block remains distinguishable from zero; the ranking of the blocks is unchanged.

Table OA.22: What the regulation index measures, states and metropolitan areas, raw and factor-adjusted networks, 1985–2025

Part 1. States

	Raw network			Factor-adjusted network		
	log TO (1)	log FROM (2)	log NET (3)	log TO (4)	log FROM (5)	log NET (6)
Panel A. Survey content and the ratings entered together						
Survey content (9)	6.41	3.00***	3.41	1.17	3.25***	−2.08
	(5.60)	(0.90)	(5.83)	(2.56)	(0.86)	(2.71)
Foster–Summers ratings (2)	5.14	−0.20	5.34	7.63*	−1.79	9.41**
	(6.47)	(1.09)	(6.68)	(3.51)	(1.43)	(3.32)
Survey minus ratings	1.26	3.20	−1.93	−6.46	5.04**	−11.49*
	(11.38)	(1.86)	(11.90)	(5.55)	(2.02)	(5.61)
	[−27.47, 25.25]	[−0.86, 7.79]	[−31.39, 22.08]	[−18.43, 6.64]	[0.42, 9.48]	[−24.40, 1.66]
Steep land share	1.09	1.02*	0.08	−1.61	0.34	−1.95
	(3.55)	(0.52)	(3.43)	(2.39)	(0.81)	(2.22)
log average distance	18.91	2.48	16.42	16.70	13.87**	2.83
	(17.74)	(3.61)	(16.89)	(13.03)	(5.34)	(11.65)
Lagged log population	12.02**	−0.26	12.29**	2.95	−3.68***	6.63**
	(4.19)	(0.63)	(4.42)	(2.23)	(0.76)	(2.35)
Wald p, blocks jointly zero	0.021	0.000	0.093	0.007	0.002	0.006
Wald p, block associations equal	0.912	0.092	0.872	0.250	0.016	0.046
Panel B. Survey content split, entered together with the ratings						
Local land-use instruments (7)	5.59	3.64***	1.96	2.09	3.26***	−1.17
	(6.41)	(0.95)	(6.72)	(3.16)	(1.12)	(3.49)
Local political pressure (1)	8.32	1.26	7.06	4.62	1.06	3.56
	(4.60)	(0.65)	(4.68)	(2.71)	(0.98)	(2.40)
Supply restrictions (1)	−2.12	−1.18	−0.94	−2.59	−0.44	−2.15
	(4.42)	(0.62)	(4.67)	(2.40)	(0.95)	(2.54)
Foster–Summers ratings (2)	1.59	−0.99	2.58	4.97	−2.20	7.17
	(6.42)	(1.20)	(6.45)	(4.23)	(1.75)	(3.84)
Wald p, blocks jointly zero	0.043	0.000	0.195	0.003	0.006	0.003
Wald p, block associations equal	0.551	0.007	0.736	0.042	0.121	0.027
Panel C. Each block in its own regression						
Survey content (9)	9.40**	2.89***	6.51*	5.60**	2.21**	3.39
	(3.54)	(0.60)	(3.44)	(2.15)	(0.81)	(2.05)
Local land-use instruments (7)	8.17*	2.81***	5.36	4.80**	2.18**	2.62
	(3.78)	(0.60)	(3.70)	(2.10)	(0.79)	(1.96)
Local political pressure (1)	10.97**	1.93**	9.04*	7.53**	1.10	6.43**
	(3.92)	(0.59)	(4.04)	(1.84)	(0.90)	(1.65)
Supply restrictions (1)	2.71	0.90	1.80	0.04	1.24	−1.20
	(4.15)	(0.71)	(4.08)	(1.98)	(0.75)	(1.94)
Foster–Summers ratings (2)	9.29*	1.74*	7.55	8.38***	0.31	8.07***
	(4.30)	(0.85)	(4.23)	(2.72)	(1.31)	(2.43)
Year fixed effects	✓	✓	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968	1,968	1,968

Note: Part 1: 48 states, 41 annual networks 1985–2025; columns (1)–(3) raw, (4)–(6) factor-adjusted. Each column is the regression of Table 2 with WRLURI 2006 replaced by blocks of its components, each block applying the index's own component weights to the equal-jurisdiction state means of its components and standardized across the 48 states; the number after a block counts its components. Survey content is the local land-use instruments, local political pressure and supply restrictions, and the two Foster–Summers ratings, the State Political Involvement and State Court Involvement Indexes, are external ratings of state legislatures and courts. Panel C enters each block in its own regression. The survey-minus-ratings row is the difference between the two block coefficients of Panel A, with the 95-percent interval that inverts the wild bootstrap test in brackets. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws). The Wald tests use the cluster-robust covariance with an $F(q, 47)$ reference distribution.

Table OA.22 (continued). Part 2, metropolitan areas

	Blocks entered together			Each block alone	
	(1)	(2)	(3)	(4)	(5)
Panel A. Raw network					
Survey content (9 components)	0.75**			0.78***	0.37***
	(0.25)			(0.26)	(0.10)
Local land-use instruments (7)		0.30	0.25	0.71**	0.38***
		(0.33)	(0.13)	(0.29)	(0.11)
Local political pressure (1)		0.64**	0.19	0.77***	0.29***
		(0.24)	(0.12)	(0.22)	(0.10)
Supply restrictions (1)		−0.06	0.01	0.04	0.04
		(0.22)	(0.06)	(0.20)	(0.08)
Foster–Summers ratings (2 components)	0.17	0.15		0.38	–
	(0.35)	(0.35)		(0.36)	
Survey minus ratings	0.58				
	(0.45)				
Wald p, blocks jointly zero	0.011	0.012	0.007	–	–
Wald p, block associations equal	0.199	0.178	0.109	–	–
Within R^2	0.076	0.081	0.048	–	–
Panel B. Factor-adjusted network					
Survey content (9 components)	0.72			0.70	0.33
	(0.61)			(0.59)	(0.22)
Local land-use instruments (7)		0.40	0.19	0.68	0.30
		(0.68)	(0.28)	(0.62)	(0.25)
Local political pressure (1)		0.68	0.13	0.78*	0.22
		(0.42)	(0.17)	(0.39)	(0.14)
Supply restrictions (1)		−0.40	0.13	−0.25	0.15
		(0.52)	(0.10)	(0.52)	(0.10)
Foster–Summers ratings (2 components)	−0.09	−0.20		0.11	–
	(0.68)	(0.71)		(0.67)	
Survey minus ratings	0.81				
	(1.03)				
Wald p, blocks jointly zero	0.494	0.262	0.315	–	–
Wald p, block associations equal	0.433	0.390	0.964	–	–
Within R^2	0.034	0.039	0.010	–	–
Controls of Table OA.12	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓
Census-division fixed effects	✓	✓	–	✓	–
Principal-state fixed effects	–	–	✓	–	✓
Metropolitan areas	100	100	64	100	64
Largest variance inflation factor	2.40	2.49	2.84	–	–

Note: Part 2: the 100 largest metropolitan areas, 41 annual networks 1985–2025, in the design of Table OA.12, column 2 (columns 1, 2 and 4) or column 3 (columns 3 and 5); Panel A raw, Panel B factor-adjusted. The dependent variable is the log of external connectedness (FROM), and every column controls for the steep land share, lagged log population and the outside-node-set share. Each block applies WRLURI's own component weights to the CBSA means of its components and is standardized across the 100 areas, so that the blocks of a column add up to the published index. The ratings are state-level and are absorbed by principal-state effects, so column 3 enters the survey blocks alone; columns 4 and 5 report each block in its own regression. The survey-minus-ratings row is the difference between the two column 1 coefficients, with a 95-percent interval of $[-0.38, 1.52]$ raw and $[-1.46, 2.99]$ factor-adjusted. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws). The Wald tests use the cluster-robust covariance with an F reference distribution.

OA.8.3 Influence of states and regions, and spatially robust inference

Regulation, physical constraint and connectedness are unevenly distributed across Census regions, so an association estimated on 48 states could rest on the contrast between a few of them. Over the 21 endpoints from 2005, mean WRLURI 2006 is −0.430 in the South, −0.562 in the Midwest, 0.785 in the Northeast and 0.040 in the West. The influence exercise of Table OA.23, Part 1, asks which states and which regions the estimate depends on. The regressions of Table 2, for each of its three outcomes, are estimated under the benchmark network with and without Census-region effects and under the own-lags-free network, and then re-estimated after removing each of the 48 targets in turn and each of the four Census regions. Outcomes and covariate scaling are unchanged throughout; omitting a target does not re-estimate a smaller national VAR or renormalize the surviving network.

In the raw network every single-state omission retains a positive WRLURI coefficient for every outcome in every model, and the state-omission ranges stay close to the corresponding full-sample estimates, at 2.52 to 3.09 percent for external connectedness under the benchmark. Removing an entire region moves the estimates considerably further. For external connectedness the minimum comes from omitting the South in the benchmark rows and from omitting the Midwest under the own-lags-free network, and with region effects the South's omission leaves only 0.17 percent; for log TO and log NET the region-controlled estimates are smallest when the Northeast is omitted and remain well above zero. That no single target produces the positive sign is not the same claim as every target carrying the same association. The reported intervals are ranges of point estimates, not confidence intervals and not a correction for residual spatial dependence; spatially robust precision is not established by deletion.

Part 2 of the table reports that precision for the sending-side cells of Table 2. The state-clustered inference of the paper allows arbitrary dependence within a state over time but none across states, although regulation is spatially clustered. The first alternative clusters the wild bootstrap by the nine Census divisions, which allows arbitrary dependence among the states of a division. The second computes Conley standard errors with a Bartlett kernel in the great-circle distance between state population centers, at cutoffs of 500, 1,000 and 1,500 kilometers, allowing correlation within a state at any lag and between nearby states at any lag. The factor-adjusted TO coefficient keeps an interval that excludes zero under every procedure, and its widest interval is the division-level one. The Conley standard errors are smaller than the state-clustered ones, because the cross-state terms they add are negative on net.

Table OA.23: The regulation coefficient under state and region exclusions, and under spatially robust inference, raw and factor-adjusted networks, 1985–2025

Part 1. Omitting each state and each Census region

		Omitting each state		Omitting one Census region				
	Full sample	Range	Minimum	Northeast	Midwest	South	West	Sign changes
Panel A. Raw network, log *TO*								
Benchmark	10.54	[9.05, 11.83]	LA	16.00	10.01	8.59	13.79	0/0
Census-region effects	21.06	[16.62, 23.38]	NY	14.06	23.86	21.42	22.02	0/0
Own-lags-free network	13.73	[12.72, 15.34]	ND	17.95	12.17	10.72	19.36	0/0
Panel B. Raw network, log *FROM*								
Benchmark	2.76	[2.52, 3.09]	LA	3.22	2.67	1.91	3.04	0/0
Census-region effects	2.17	[1.49, 2.79]	MD	2.93	2.29	0.17	2.72	0/0
Own-lags-free network	5.52	[4.99, 6.00]	NV	5.85	4.82	5.62	6.04	0/0
Panel C. Raw network, log *NET*								
Benchmark	7.78	[6.50, 9.07]	CO	12.78	7.34	6.68	10.75	0/0
Census-region effects	18.89	[13.83, 21.29]	NY	11.13	21.56	21.25	19.30	0/0
Own-lags-free network	8.21	[6.79, 9.44]	RI	12.11	7.36	5.11	13.32	0/0
Panel D. Factor-adjusted network, log *TO*								
Benchmark	7.50	[6.38, 8.57]	LA	11.15	7.00	6.71	10.42	0/0
Census-region effects	11.46	[9.69, 13.14]	NY	9.76	11.04	13.56	11.37	0/0
Own-lags-free network	9.14	[8.50, 10.18]	LA	10.78	8.18	8.74	10.68	0/0
Panel E. Factor-adjusted network, log *FROM*								
Benchmark	1.67	[0.92, 2.01]	LA	2.72	1.71	0.36	1.94	0/0
Census-region effects	1.92	[0.97, 2.59]	LA	2.11	2.24	0.15	2.14	0/0
Own-lags-free network	4.95	[4.39, 5.47]	NH	3.72	4.99	4.92	4.88	0/0
Panel F. Factor-adjusted network, log *NET*								
Benchmark	5.83	[4.99, 7.18]	RI	8.43	5.29	6.34	8.47	0/0
Census-region effects	9.55	[7.15, 10.73]	NY	7.65	8.81	13.40	9.23	0/0
Own-lags-free network	4.18	[3.71, 5.01]	CO	7.06	3.19	3.82	5.80	0/0

Note: Part 1: 48 states, 41 annual networks 1985–2025. Each entry is 100 times the WRLURI 2006 coefficient of Table 2 under the named network and model; the benchmark rows are that table, the region-effects rows add Census-region effects and the own-lags-free rows use the network whose own lags are unpenalized. Each model is estimated on the full sample, after removing each of the 48 states in turn and after removing each Census region in turn, without re-estimating the national VAR or renormalizing the surviving network and with the full-sample standardizations. The range and the state under Minimum refer to the 48 state omissions; ranges are ranges of point estimates, not confidence intervals. Sign changes count the omissions whose estimate has the opposite sign to the full-sample estimate.

Table OA.23 (continued). Part 2, clustered and Conley inference for the sending side

		Clustered, wild bootstrap-t		Conley, Bartlett kernel		
	Estimate	48 states	9 divisions	500 km	1,000 km	1,500 km
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Factor-adjusted network, columns (4) and (6) of Table 2						
log TO	7.50	(2.32)	(2.08)	(2.07)	(2.09)	(2.00)
		[2.67, 12.70]	[1.15, 16.38]	[3.34, 11.66]	[3.29, 11.71]	[3.48, 11.52]
		{0.004}	{0.031}	{< 0.001}	{< 0.001}	{< 0.001}
log NET	5.83	(2.08)	(1.41)	(1.72)	(1.62)	(1.44)
		[1.40, 10.43]	[3.03, 11.87]	[2.38, 9.28]	[2.57, 9.09]	[2.94, 8.72]
		{0.012}	{0.012}	{0.001}	{< 0.001}	{< 0.001}
Panel B. Raw network, columns (1) and (3) of Table 2						
log TO	10.54	(3.62)	(3.07)	(3.18)	(2.92)	(2.69)
		[2.49, 18.06]	[0.59, 20.94]	[4.14, 16.93]	[4.66, 16.41]	[5.12, 15.95]
		{0.017}	{0.045}	{0.002}	{< 0.001}	{< 0.001}
log NET	7.78	(3.40)	(2.66)	(3.04)	(2.69)	(2.45)
		[0.15, 14.82]	[−1.61, 16.18]	[1.66, 13.90]	[2.37, 13.19]	[2.86, 12.70]
		{0.047}	{0.072}	{0.014}	{0.006}	{0.003}

Note: Part 2: 48 states, 41 annual networks 1985–2025. Each row is the WRLURI 2006 coefficient of the named column of Table 2, 100 times the coefficient, with standard errors in parentheses, 95-percent intervals in brackets and two-sided p-values in braces. Column (2) repeats the inference of that table, a cluster-robust standard error by state with the p-value and interval of a wild cluster bootstrap-t that imposes the null (9,999 draws of Webb's weights). Column (3) applies the same procedure with the nine Census divisions as clusters. Columns (4) to (6) report Conley standard errors, which allow the errors to be correlated within a state across all endpoints and between two states within the cutoff at any lag, with Bartlett weight $\max(0, 1 - d/\bar{d})$ on the great-circle distance d between the states' population centers; they are computed after the year effects are partialled out, carry no small-sample factor and use Student's t with 47 degrees of freedom. On average a state has 4.3, 13.9 and 23.6 other states within 500, 1,000 and 1,500 km. The division and Conley standard errors are below the state-clustered one in every row because the cross-state terms they add are negative on net, partly mechanically, since the year effects make the 48 states' score contributions sum to zero.

OA.8.4 Net positions by Census region

Table OA.24 describes the raw benchmark network's net positions by Census region and for each Northeast state. The comparison sets the paper beside Gabauer et al. (2024), whose Diebold–Yilmaz network of state house prices has the Northeast receiving shocks from the other regions. Over every stable monthly window, in the raw network the Northeast is on average a net receiver, at −6.2 percentage points over 1985–2025 and −16.6 from 2005, and New York is its largest net receiver, below zero in 91 percent of the windows and in all but one of those ending from 2005. The most regulated Northeast states, Massachusetts, Rhode Island and New Hampshire, are net sources on average. The regulation gradient of Section 5 is estimated among states within the same broad region as well as across regions.

Table OA.24: Net position by Census region and Northeast state, raw benchmark network

		1985–2025		2005–2025	
	WRLURI 2006	Mean	Below zero (%)	Mean	Below zero (%)
Census regions					
Northeast		−6.2	57.7	−16.6	69.0
Midwest		−6.2	64.2	−6.9	65.1
South		2.6	53.5	2.6	55.8
West		8.0	51.2	17.4	49.6
Northeast states, by WRLURI 2006					
MA	2.39	7.7	46.0	12.0	46.4
RI	2.32	10.7	29.6	7.8	36.8
NH	2.11	18.2	45.2	13.0	61.2
NJ	1.43	−2.3	47.7	−26.7	76.4
ME	1.11	−11.7	70.4	−10.5	64.8
CT	0.71	16.5	30.4	9.0	44.8
PA	0.70	−21.9	71.0	−47.1	100.0
VT	0.66	−32.7	87.9	−45.1	90.8
NY	0.15	−40.1	90.6	−62.1	99.6

Note: Raw benchmark network of the 48 states, every dynamically stable 120-month window ending from July 1985 through December 2025, and those ending from January 2005. The net difference is TO minus FROM in percentage points of forecast-error variance, and a state with a negative value is a net receiver. Mean is the mean net difference and Below zero the percentage of windows in which it is negative. Regional rows average over the state-months of the region, and their percentages are of state-months. WRLURI 2006 is standardized across the 48 states. The table describes the raw network only, the network closest in design to that of Gabauer et al. (2024), and involves no second-stage regression.

OA.9 Spatial-Equilibrium Derivations

The connectedness estimator does not impose a migration or distance matrix on the dynamics of house-price growth. A theory-based representation therefore requires a separate object that has the same directed source–target dimension. The location-choice model in Schubert (2024) provides one. It maps household movements into a row-stochastic migration exposure matrix and then into equilibrium price weights. This appendix derives that mapping, establishes the relevant cross-pair and target-side comparative statics, and shows how the direct weights aggregate to target, source, and system objects. The table lists the symbols of the model, each with one meaning, and the four empirical objects that correspond to them.

Symbol	Meaning
A_{it}	Amenity level of location i.
a_{it}	Log change in the amenity level.
E_{it}	Wage in location i.
e_{it}	Log change in the wage.
P_{it}	House price, the cost of housing in location i.
p_{it}	Log change in the house price.
L_{it}	Population of location i.
l_{it}	Log change in population.
α	Housing expenditure share.
ν	Dispersion of the idiosyncratic location taste.
κ_i	Housing-supply shifter.
η_i^H	Inverse housing-supply elasticity of location i.
$\mu_t^{k\to j}$	Share of origin-k households choosing destination j.
$\varphi_t^{i\leftarrow k}$	Share of target i's feeder population supplied by origin k.
$\psi_{ij,t}$	Migration exposure of target i to source j; $\psi_{ij,t}^0$ is the pre-window matrix of the empirical design.
$o_{i,t}$	External migration openness, $1-\psi_{ii,t}$.
c	The ratio α/ν.
a_i	The constant $c\,\eta_i^H$ of target i.
$\chi_{i,t}$	The product $a_i o_{i,t}$.
$\rho_{i,t}$	Supply multiplier $a_i/(1+a_i o_{i,t})$.
$\eta_{i,t}^{nw}$	Target i's total external weight $\chi_{i,t}/(1+\chi_{i,t})$.
$W_{ij,t}$	Direct equilibrium price weight from source j to target i.
$Q_{ki,t}$	Pooled IRS migration flow from origin k to destination i.
$F_{i,t}$	Measured external connectedness of target i.
$\bar{C}_{ij,t}$	Conditional connectedness share of source j in target i's row.
$x_{ij,t}$	Conditional migration exposure, the centered regressor of Section 4.

OA.9.1 Household location choice and spatial equilibrium

Let $i, j, k \in \{1, \ldots, N\}$ index locations. A worker ω residing in location i receives indirect utility

$$V_{it}(\omega) = \log A_{it} + \log E_{it} - \alpha \log P_{it} + z_i(\omega) \equiv U_{it} + z_i(\omega), \tag{OA.4}$$

where A_{it} is the amenity level, E_{it} the wage, P_{it} the cost of housing or local nontradables, and $\alpha > 0$ the housing expenditure weight. The idiosyncratic taste $z_i(\omega)$ follows a Type-I extreme-value distribution with dispersion $\nu > 0$.

A worker initially in k chooses the destination that maximizes $U_{it} - \tau^{ki} + z_i(\omega)$, where τ^{ki} is an additive, time-invariant cost of moving from k to i. The resulting share of workers from origin k who choose destination i is

$$\mu_t^{k \to i} = \frac{\exp\Big[(U_{it} - \tau^{ki})/\nu\Big]}{\sum_{r=1}^{N} \exp[(U_{rt} - \tau^{kr})/\nu]}. \tag{OA.5}$$

The superscript lists the origin first and the destination second. The inverse dispersion ν^{-1} determines the sensitivity of destination shares to common utility differences. Persistent bilateral moving costs are therefore consistent with a heterogeneous migration network even when local fundamentals change over time.

The inverse supply curve links housing prices to population:

$$P_{it} = \kappa_i L_{it}^{\eta_i^H} \exp(\varepsilon_{it}^H), \tag{OA.6}$$

where L_{it} is population, κ_i is a time-invariant supply shifter, ε_{it}^H is a local supply disturbance, and $\eta_i^H > 0$ is the inverse housing-supply elasticity. A larger η_i^H denotes a steeper inverse relation between population and price. Destination populations satisfy

$$L_{it} = \sum_{k=1}^{N} \mu_t^{k \to i} L_{k,t-1}. \tag{OA.7}$$

An equilibrium at date t is a pair (P_t, L_t) satisfying equations (OA.5)–(OA.7), given A_t, E_t, L_{t-1}, κ, ε_t^H, the bilateral moving costs, and the model parameters.

The model is deliberately spare. Workers are myopic, wages are exogenous to this block, and P_{it} is both the housing-services cost entering utility and the observed house price. There is no separate rent, user-cost, tenure, or mortgage sector. Allowing η_i^H to vary across locations implements the heterogeneous-supply extension of Schubert's common-elasticity benchmark.

OA.9.2 Log-linear reallocation and the migration exposure matrix

Lowercase letters denote log changes. In particular, $p_{it} = \Delta \log P_{it}$, $e_{it} = \Delta \log E_{it}$, and $l_{it} = \Delta \log L_{it}$. Define the change in common location utility as

$$q_{it} = a_{it} + e_{it} - \alpha p_{it}. \tag{OA.8}$$

Log differentiation of the destination share gives

$$\Delta \log \mu_t^{k \to i} = \nu^{-1} \left[q_{it} - \sum_{j=1}^{N} \mu_{t-1}^{k \to j} q_{jt} \right]. \tag{OA.9}$$

An origin therefore sends a larger share to i when that destination becomes more attractive relative to the other locations available to the same households.

To aggregate these choices at the target, define the share of destination i's feeder population originating in k:

$$\varphi_{t-1}^{i \leftarrow k} = \frac{\mu_{t-1}^{k \to i} L_{k,t-2}}{\sum_{r=1}^{N} \mu_{t-1}^{r \to i} L_{r,t-2}} = \frac{\mu_{t-1}^{k \to i} L_{k,t-2}}{L_{i,t-1}}. \tag{OA.10}$$

Log differentiation of population accounting yields

$$l_{it} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} \left(\Delta \log \mu_t^{k \to i} + l_{k,t-1} \right). \tag{OA.11}$$

Substituting the change in destination shares and collecting terms gives

$$l_{it} = \Delta M_{i,t-1} + \nu^{-1} \left[(a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it}) - \alpha (p_{it} - \widetilde{p}_{it}) \right], \tag{OA.12}$$

where

$$\Delta M_{i,t-1} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} l_{k,t-1} \tag{OA.13}$$

captures growth in the populations feeding i, and the two-step network average of any log change v is

$$\widetilde{v}_{it} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} \sum_{j=1}^{N} \mu_{t-1}^{k \to j} v_{jt}. \tag{OA.14}$$

The remaining terms in equation (OA.12) compare amenities, wages, and housing costs in target i with the alternatives considered by its feeder origins.

The two-step average defines the migration exposure matrix:

$$\psi_{ij,t-1} = \sum_{k=1}^{N} \varphi_{t-1}^{i \leftarrow k} \mu_{t-1}^{k \to j}, \qquad \sum_{j=1}^{N} \psi_{ij,t-1} = 1. \tag{OA.15}$$

Exposure from source j to target i is large when origins that contribute heavily to i also assign a large destination share to j. It therefore measures competition for common migrant pools rather than only the direct flow from j to i. In the state data, with stayers on the diagonal, the direct terms of equation (8) dominate, and the common-origin term is a small share of external exposure. Using this definition, the two-step average becomes

$$\widetilde{v}_{it} = \sum_{j=1}^{N} \psi_{ij,t-1} v_{jt}. \tag{OA.16}$$

The diagonal ψ_{ii} is own exposure. Its complement $o_i \equiv 1 - \psi_{ii}$ is the target's external migration openness. For $j \neq i$, define the composition of that external exposure and the corresponding network house-price change as

$$\widetilde{\psi}_{ij,t-1} = \frac{\psi_{ij,t-1}}{1 - \psi_{ii,t-1}}, \qquad \sum_{j \neq i} \widetilde{\psi}_{ij,t-1} = 1, \tag{OA.17}$$

$$p_{it}^{nw} = \sum_{j \neq i} \widetilde{\psi}_{ij,t-1} p_{jt}. \tag{OA.18}$$

The conditional weights $\widetilde{\psi}_{ij}$ are useful for deriving the external total. The diagonal-inclusive ψ_{ij} underlies openness, and the main conditional-allocation estimand uses $\widetilde{\psi}_{ij}$.

OA.9.3 Collecting the own-price terms

Log differentiation of inverse housing supply gives

$$p_{it} = \eta_i^H l_{it} + \Delta\varepsilon_{it}^H. \tag{OA.19}$$

Substituting population reallocation, applying the exposure average, and separating target i's own price gives

$$\begin{aligned} p_{it} = {} & \eta_i^H \Delta M_{i,t-1} + \eta_i^H \nu^{-1} \left[(a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it}) \right] \\ & - \alpha \eta_i^H \nu^{-1} p_{it} + \alpha \eta_i^H \nu^{-1} \psi_{ii,t-1} p_{it} + \alpha \eta_i^H \nu^{-1} \sum_{j \neq i} \psi_{ij,t-1} p_{jt} + \Delta\varepsilon_{it}^H. \end{aligned} \tag{OA.20}$$

Moving both own-price terms to the left gives

$$\begin{aligned} & \left[1 + \alpha \nu^{-1} \eta_i^H (1 - \psi_{ii,t-1}) \right] p_{it} \\ & \quad = \alpha \nu^{-1} \eta_i^H \sum_{j \neq i} \psi_{ij,t-1} p_{jt} + \eta_i^H \Delta M_{i,t-1} \\ & \qquad + \eta_i^H \nu^{-1} \left[(a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it}) \right] + \Delta\varepsilon_{it}^H. \end{aligned} \tag{OA.21}$$

Define

$$\zeta_{i,t-1} = \frac{\eta_i^H}{1 + \alpha \nu^{-1} \eta_i^H (1 - \psi_{ii,t-1})}. \tag{OA.22}$$

The time-varying solution within this log-linearized system is then

$$p_{it} = \alpha\nu^{-1}\zeta_{i,t-1}\sum_{j\neq i}\psi_{ij,t-1}p_{jt} + \xi_{it}, \tag{OA.23}$$

with non-network component

$$\xi_{it} = \zeta_{i,t-1}\Delta M_{i,t-1} + \zeta_{i,t-1}\nu^{-1}\left[(a_{it} - \widetilde{a}_{it}) + (e_{it} - \widetilde{e}_{it})\right] + \frac{\zeta_{i,t-1}}{\eta_i^H}\Delta\varepsilon_{it}^H. \tag{OA.24}$$

The empirical implementation associates each annual network t with an exposure matrix Ψ_t^0 measured before its house-price window. The matrix is fixed within that window but advances from one year's network to the next. Define

$$\zeta_{i,t}^0 = \frac{\eta_i^H}{1 + \alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)}, \qquad \eta_{i,t}^{nw} = \alpha\nu^{-1}\zeta_{i,t}^0(1-\psi_{ii,t}^0). \tag{OA.25}$$

Equivalently, target i's total external weight is

$$\eta_{i,t}^{nw} = \frac{\alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)}{1 + \alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)} \in (0,1). \tag{OA.26}$$

Using $\widetilde{\psi}_{ij,t}^0 = \psi_{ij,t}^0/(1-\psi_{ii,t}^0)$, the price equation for date s inside window t becomes

$$p_{i,s} = \eta_{i,t}^{nw}\sum_{j\neq i}\widetilde{\psi}_{ij,t}^0 p_{j,s} + \varepsilon_{i,s}^{nw} = \eta_{i,t}^{nw}p_{i,s}^{nw} + \varepsilon_{i,s}^{nw}. \tag{OA.27}$$

The residual retains the parts of the exact solution not represented by the predetermined network-price term:

$$\begin{aligned}\varepsilon_{i,s}^{nw} = {} & \underbrace{\zeta_{i,s-1}\Delta M_{i,s-1}}_{\text{migration access}} + \underbrace{\zeta_{i,s-1}\nu^{-1}\left[(a_{i,s} - \widetilde{a}_{i,s}) + (e_{i,s} - \widetilde{e}_{i,s})\right]}_{\text{relative amenities and wages}} \\ & \underbrace{+ \alpha\nu^{-1}\sum_{j\neq i}\left(\zeta_{i,s-1}\psi_{ij,s-1} - \zeta_{i,t}^0\psi_{ij,t}^0\right)p_{j,s}}_{\text{predetermined-weight approximation}} + \underbrace{\frac{\zeta_{i,s-1}}{\eta_i^H}\Delta\varepsilon_{i,s}^H}_{\text{local supply disturbance}}.\end{aligned} \tag{OA.28}$$

The approximation term is current minus predetermined because equation (OA.27) replaces the exact coefficient $\zeta_{i,s-1}\psi_{ij,s-1}$ with $\zeta_{i,t}^0\psi_{ij,t}^0$.[1]

Schubert's empirical baseline holds a single 1990–1995 matrix fixed across all outcome years. The algebra requires only that the weights be determined before the outcome window they summarize. The present design implements that timing with 21 advancing pre-window matrices, as described in Section 4.1.

[1]The corresponding approximation in Schubert (2024) presents the difference in the reverse order. Substitution into the preceding derivation yields the current-minus-predetermined order used here.

Let $\chi_{i,t} = \alpha\nu^{-1}\eta_i^H(1-\psi_{ii,t}^0)$. Because $\eta_{i,t}^{nw} = \chi_{i,t}/(1+\chi_{i,t})$,

$$\frac{\partial \eta_{i,t}^{nw}}{\partial(1-\psi_{ii,t}^0)} = \frac{\alpha\nu^{-1}\eta_i^H}{(1+\chi_{i,t})^2} > 0, \qquad \frac{\partial \eta_{i,t}^{nw}}{\partial \eta_i^H} = \frac{\alpha\nu^{-1}(1-\psi_{ii,t}^0)}{(1+\chi_{i,t})^2} > 0. \tag{OA.29}$$

The algebraic external total is larger for a more migration-open target and a target with less elastic housing supply. It approaches zero when either openness or the inverse supply elasticity approaches zero, and it approaches one as η_i^H or ν^{-1} becomes arbitrarily large.

OA.9.4 Model relationships used for the empirical comparison

Set $c \equiv \alpha\nu^{-1}$, $a_i \equiv c\eta_i^H$, and $o_{i,t} \equiv 1-\psi_{ii,t}^0$. For a particular external link, the price equation implies

$$W_{ij,t} = \frac{a_i o_{i,t}}{1+a_i o_{i,t}} \frac{\psi_{ij,t}^0}{o_{i,t}} = \underbrace{\frac{a_i}{1+a_i o_{i,t}}}_{\rho_{i,t}} \psi_{ij,t}^0, \qquad j \neq i.$$

The first representation separates the external total from the composition of the external row. The second uses the diagonal-inclusive exposure retained for openness.

Within a target row, openness $o_{i,t}$ is fixed across sources. Holding it fixed while exposure is reallocated among external links gives

$$\left.\frac{\partial W_{ij,t}}{\partial \psi_{ij,t}^0}\right|_{o_{i,t}} = \rho_{i,t} > 0, \qquad \left.\frac{\partial W_{ij,t}}{\partial \eta_i^H}\right|_{\psi_{ij,t}^0, o_{i,t}} = \frac{c\psi_{ij,t}^0}{(1+a_i o_{i,t})^2} > 0, \qquad \frac{\partial \rho_{i,t}}{\partial \eta_i^H} = \frac{c}{(1+a_i o_{i,t})^2} > 0.$$

Greater diagonal-inclusive exposure therefore receives a larger direct weight. The level gradient between exposure and that weight is also steeper when target supply is more constrained. The model log elasticity, by contrast, is one:

$$\left.\frac{\partial \log W_{ij,t}}{\partial \log \psi_{ij,t}^0}\right|_{o_{i,t}} = 1.$$

An alternative row-stochastic comparison adds mass to link ij entirely by reducing the diagonal, so $do_{i,t}/d\psi_{ij,t}^0 = 1$ while the remaining external links stay fixed. Then

$$\frac{dW_{ij,t}}{d\psi_{ij,t}^0} = \frac{a_i\left[1+a_i(o_{i,t}-\psi_{ij,t}^0)\right]}{(1+a_i o_{i,t})^2} > 0.$$

Monotonicity in the diagonal-inclusive link survives, although its slope differs from the fixed-openness comparison.

Openness requires a separate conditioning statement. If one diagonal-inclusive external link is held fixed while openness rises elsewhere in the row,

$$\left.\frac{\partial W_{ij,t}}{\partial o_{i,t}}\right|_{\psi^0_{ij,t}} = -\frac{a_i^2 \psi^0_{ij,t}}{(1+a_i o_{i,t})^2} < 0.$$

If every external link expands proportionately so that its composition $\psi^0_{ij,t}/o_{i,t}$ is fixed,

$$\left.\frac{dW_{ij,t}}{do_{i,t}}\right|_{\psi^0_{ij,t}/o_{i,t}} = \frac{a_i(\psi^0_{ij,t}/o_{i,t})}{(1+a_i o_{i,t})^2} > 0.$$

The unconditional openness implication applies to the target total:

$$\sum_{j\neq i} W_{ij,t} = \frac{a_i o_{i,t}}{1+a_i o_{i,t}} = \eta^{nw}_{i,t}, \qquad \frac{\partial \sum_{j\neq i} W_{ij,t}}{\partial o_{i,t}} = \frac{a_i}{(1+a_i o_{i,t})^2} > 0.$$

Equivalently, its log odds satisfy

$$\log\left(\frac{\sum_{j\neq i} W_{ij,t}}{1-\sum_{j\neq i} W_{ij,t}}\right) = \log a_i + \log o_{i,t}.$$

Lower own exposure corresponds to a larger target external total. Every pairwise weight shares that ordering only along a proportional expansion that holds the composition of external exposure fixed.

The model's comparative statics therefore concern two objects, target external connectedness and conditional source allocation. The GFEVD row total is a reduced-form counterpart to η^{nw}, and off-diagonal row shares describe the allocation ordering in $\widetilde{\Psi}$. Bilateral levels combine both margins. The paper's sending-side result is not among these objects. A source's total outward weight, equation (OA.35), depends on the other markets' constraints and exposures rather than on the source's own constraint, and the model does not derive how a source's constraint moves its share of other markets' forecast-error variance, so the sending-side reading is an informal extension of the model. Dynamics and innovation covariances keep all measured GFEVD objects distinct from the model weights.

OA.9.5 Microfoundation of the weighting matrix

The choice probabilities make the direction of the exposure weight transparent. For origin k, differentiating the probability of choosing destination r with respect to common utility in destination j gives

$$\frac{\partial \mu_t^{k\to r}}{\partial q_{jt}} = \nu^{-1}\mu_t^{k\to r}\left(\mathbf{1}\{r=j\} - \mu_t^{k\to j}\right). \tag{OA.30}$$

The derivatives are positive for a destination's own utility and negative for an alternative destination's utility. Aggregating them with the feeder weights in Φ_t yields

$$\frac{\partial(l_{it} - \Delta M_{i,t-1})}{\partial q_{jt}} = \begin{cases} \nu^{-1}(1 - \psi_{ii,t-1}), & j = i, \\ -\nu^{-1}\psi_{ij,t-1}, & j \neq i. \end{cases} \tag{OA.31}$$

Because q_{jt} contains $-\alpha p_{jt}$, the corresponding price derivatives are

$$\frac{\partial(l_{it} - \Delta M_{i,t-1})}{\partial p_{jt}} = \begin{cases} -\alpha\nu^{-1}(1 - \psi_{ii,t-1}), & j = i, \\ \alpha\nu^{-1}\psi_{ij,t-1}, & j \neq i. \end{cases} \tag{OA.32}$$

These derivatives encode the model's allocation comparison. The cross-destination term is proportional to ψ_{ij}, and the associated inverse-supply factor varies with the target's constraint parameter.

Using $\rho_{i,t}$ from Section OA.9.4, the direct off-diagonal model weight can be written either with conditional external exposure or with the raw empirical exposure:

$$W_{ij,t} = \eta^{nw}_{i,t}\widetilde{\psi}^{0}_{ij,t} = \rho_{i,t}\psi^{0}_{ij,t}, \qquad i \neq j. \tag{OA.33}$$

The two expressions use different normalizations. $\widetilde{\psi}^{0}_{ij,t}$ allocates exposure conditional on choosing an external destination, whereas $\psi^{0}_{ij,t}$ is the full-row weight used for openness. The main conditional allocation uses $\widetilde{\psi}^{0}_{ij,t}$. Their scale factors are $\eta^{nw}_{i,t}$ and $\rho_{i,t}$, respectively.

Summing the direct weights by target, source, and system gives

$$\sum_{j \neq i} W_{ij,t} = \eta^{nw}_{i,t}, \tag{OA.34}$$

$$\sum_{i \neq j} W_{ij,t} = \sum_{i \neq j} \rho_{i,t}\psi^{0}_{ij,t}, \tag{OA.35}$$

$$\frac{1}{N}\sum_{i}\sum_{j \neq i} W_{ij,t} = \frac{1}{N}\sum_{i} \eta^{nw}_{i,t}. \tag{OA.36}$$

These accounting identities connect the pairwise weights to the aggregate comparisons discussed above.

OA.9.6 System weight operator

To describe simultaneous adjustment across all locations, hold amenities and wages fixed and let $\sigma_{i,t}$ be an unexpected local housing-cost disturbance. The equilibrium price relation is

$$\widehat{p}_{i,t} = \rho_{i,t}\sum_{j \neq i} \psi^{0}_{ij,t}\widehat{p}_{j,t} + b_{i,t}\sigma_{i,t}, \qquad b_{i,t} = \frac{1}{1 + \chi_{i,t}} = 1 - \eta^{nw}_{i,t}. \tag{OA.37}$$

Define the window-specific weight and disturbance-loading matrices as

$$(W_t)_{ij} = \begin{cases} \rho_{i,t}\psi^{0}_{ij,t}, & i \neq j, \\ 0, & i = j, \end{cases} \qquad D_t = \operatorname{diag}(b_{1,t}, \ldots, b_{N,t}). \tag{OA.38}$$

Stacking these equations gives

$$\widehat{\boldsymbol{p}}_t = W_t\widehat{\boldsymbol{p}}_t + D_t\boldsymbol{\sigma}_t, \tag{OA.39}$$

and hence

$$\widehat{\boldsymbol{p}}_t = (I - W_t)^{-1} D_t \boldsymbol{\sigma}_t \equiv \mathcal{A}_t \boldsymbol{\sigma}_t. \tag{OA.40}$$

Each row of W_t sums to $\eta_{i,t}^{nw} < 1$, so its spectral radius is below one and the inverse has the Neumann-series representation

$$\mathcal{A}_t = (I - W_t)^{-1} D_t = \left(I + W_t + W_t^2 + W_t^3 + \cdots \right) D_t. \tag{OA.41}$$

The first-order term contains the direct migration routes. Higher powers collect indirect paths through intermediate markets. If local disturbances have covariance matrix $\Sigma_{\sigma,t}$, equilibrium prices have covariance

$$\Sigma_{p,t} = \mathcal{A}_t \Sigma_{\sigma,t} \mathcal{A}_t'. \tag{OA.42}$$

Theoretical exposure and empirical connectedness can share geography without being identical. Migration exposure represents cross-pair allocation, and the regressions evaluate directional comparisons without imposing identity. A generalized variance decomposition additionally depends on moving-average coefficients and innovation covariance. Connectedness can therefore change with fixed migration weights, while indirect paths can generate a large variance contribution from a small direct equilibrium edge: $C_{ij,t} \neq \psi_{ij,t}$ and $C_{ij,t} \neq W_{ij,t}$.